

Hou's Navier-Stokes singularity scenario: a resolved front in the class excluded by the Liouville theorems, and its certified centrifugal instability

[Author]
[Affiliation]

Draft 0.12, 2026-09-09 (normalization correction: every collapse-unit rate and every fraction of the remaining time in Sections 3 and 4 recomputed from the raw logs against the fitted singular time $T = 0.0022865$, the Euler branch kept on the stated convention $T_E = 0.002278$ with its band; four claims inverted; figures regenerated; threshold correction: the rate that a perturbation must exceed to outrun the Type I collapse is one half, not one, made consistent in the abstract and Sections 3.2, 3.3, 3.4, 3.6, 3.7, 7.1 and 7.5, the cascade margins and e-folds of Section 7.1 recomputed on it, and the theorems that make one half a guaranteed floor cited; provenance pass: the twelve untraceable numbers of Sections 3.1 to 4 traced to their artefacts or recomputed, the Euler Type I constant, the Euler nonlinear lags and the viscous nonlinear comparison corrected, and the amplification-label correction carried into the LaTeX, where it had not been applied). Author list, affiliations and acknowledgements to be added.

Abstract

Hou's interior scenario for singularity formation in the three-dimensional axisymmetric Euler equations, and its Navier-Stokes continuation with a two-stage viscosity, are the most developed candidates for a finite-time singularity of a smooth finite-energy incompressible flow. We continue the viscous scenario at Hou's first-stage viscosity to 331 times the initial amplitude with a front-following axial map that resolves the sharp front to ten cells (six at the last point), and measure the quantity $\sup |u| |x'|$ that is the hypothesis of the Koch-Nadirashvili-Seregin-Sverak Liouville theorem for bounded ancient axisymmetric solutions with swirl. It does not involve the fitted singular time and is constant to 0.42 percent (72.2) on the twelve snapshots we retain over 3.3 units of $\log(1/(T-t))$, from 9 to 331 times the initial swirl amplitude (one further snapshot, at $t = 0.002285549$, is excluded from this series: its map is degraded (Section 5.2) and it reads 556, so the series as quoted omits its largest value and the constancy claim is made on the retained points only); its value is carried by the outer flow, and at the swirl maximum itself the same quantity is flat at 21 to 22 where a logarithmic growth would read thirty percent. Over the resolved range the scenario as computed therefore satisfies the hypothesis under which the exclusion theorems apply, which is the opposite of the logarithmic escape Hou identified in his generalized model at far larger amplifications; a finite computation cannot establish the bound up to T , and on the core-restricted ru^r the data are inconclusive (non-monotone between 15 and 22 over the range). The front is the Rott-Lundgren strained layer under the measured collapse strain to a five percent profile residual, the amplitude is Type I (exponent 1.03 ± 0.03) with a converged singular time ($0.0022865 \pm 1 \times 10^{-7}$) by a log-derivative fit, the meridional Type I constant is bounded, and the scale-invariant local L^3 norm at the collapse scale decreases over four units of $\log(1/(T-t))$. Continued past that range the solution leaves the one-scale law: at about 400 times the initial amplitude the primary front stalls, a vacuum region opens between it and the axis, and a second structure nearer the symmetry plane takes over, after which the

amplitude pulses rather than following a power law. Followed further at ten times denser output it does not arrest but reaches between four and six thousand times the initial amplitude, passing the fitted singular time, until the structure is too thin to resolve on any grid we can run; its growth over that stretch is punctuated by the same pulsing the transition introduced. Through all of it the circulation stays pinned by its maximum principle and the local L^3 norm stays flat, so neither quantity that must diverge at a singularity does. We also compute, for the first time, the linear stability of the interior collapse to non-axisymmetric perturbations, on the frozen and on the co-evolving base, in Euler and in Navier-Stokes at both of Hou’s viscosities, with resolution certificates in both grid directions. Collapse units on the viscous states use the fitted singular time 0.0022865 of the amplitude fit below; the Euler states keep Hou’s checkpoint 0.002278 as an explicitly stated convention, because no Euler singular-time fit converges, with a systematic band of about nine percent attached, and the Euler conclusions rest on ratios, which do not depend on that choice. The threshold a perturbation must clear to outrun a Type I collapse is half the collapse rate, and that is also the floor the instability theorems of Vasseur and Vishik [VV20] and Lafleche, Vasseur and Vishik [LVV21] already guarantee, so what the table contributes is not that the bar is cleared but which families clear it and by how much. On the Euler state the long-wave axisymmetric direction is marginally stable (peak 0.495 of the collapse rate at three resolutions, just below the threshold); on the viscous state the long-wave $m = 1$ direction is not held below the threshold by either of Hou’s viscosities, crossing it at 10.7 percent of the remaining time at 5×10^{-4} , passing the collapse rate itself at 27.2 percent and reaching 1.18, or 2.4 times the threshold, at 34 percent. The short-wave shearing continuum of the front is unstable relative to the collapse at every azimuthal wavenumber tested ($m = 0, 1, 2$), crossing half the collapse rate at 4.2 to 4.7 percent of the remaining time at viscosity 5×10^{-4} and at 14 to 19 percent at 5×10^{-3} , and passing the collapse rate itself at 6.9 to 15.2 percent at 5×10^{-4} , so that at the last resolved point the three stand at 1.48, 2.15 and 2.08, three to four times the threshold; none of the three saturates within the resolved range; the crossing fraction of the $m = 1$ family rises roughly logarithmically with viscosity from 5×10^{-4} to 1×10^{-2} . Non-linearly, no seed, state or viscosity tested transfers the collapse to the symmetry-broken state; on the Euler states the feedback depletes it, by 22 percent in amplitude by half the remaining time, while on the viscous state the amplitude is not measurably depleted within the range we resolve. The instability is identified as the centrifugal instability of the front, whose circulation decreases outward, extended to every azimuthal wavenumber by the axial flow of the collapse: a one-dimensional parallel-flow model of the front column has an isolated unstable eigenvalue whose inviscid rate approaches the local Rayleigh bound (1.47 of the collapse rate) from below and whose viscous peak (1.24) brackets the two-dimensional frozen-base rate (1.21 to 1.24; the model is local, its rate spans 0.78 to 1.61 across the front’s thickness and it fails on the inviscid state), and that eigenvalue is certified by a Krawczyk enclosure with rigorous floating-point bounds for the collocation discretization of the model on an explicit rational polynomial field, at three resolutions, the two finest agreeing to nine digits; the certified rate is 1.2447 in collapse units at Hou’s first-stage viscosity and 1.0080, 1.2999 and 1.2081 at his second stage, inviscid and at $m = 0$, so what is certified is not only that the front column is unstable but that this identified direction outruns the collapse by a factor of 2.0 to 2.6 over the threshold, the margin being smallest at Hou’s second-stage viscosity, where the claim is correspondingly weakest (the enclosures are statements about the eigenvalue in inverse seconds and do not involve the singular time; only the conversion to collapse units does); the Leibovich-Stewartson sufficient condition is certified on the same kind of field by Sturm root counting in exact arithmetic. The local front structure is an exact infinite-energy Navier-Stokes solution, which explains why a core-resolving computation cannot observe the departure that finite energy forces at a few collapse lengths.

1 Introduction

The regularity problem for the three-dimensional incompressible Navier-Stokes equations remains open in both directions. On the numerical side the strongest evidence for singular behaviour comes

from the axisymmetric flows with swirl of Luo and Hou [LH14a, LH14b] at a solid boundary and, in the interior, of Hou [H23a] for Euler and Hou [H23b] for Navier-Stokes. The boundary scenario now has a computer-assisted proof of stable nearly self-similar blowup for Euler [CH25]; the interior scenario is supported by a self-similar profile fitted through amplifications of order 10^7 in Euler and, with a viscosity of 5×10^{-4} raised to 5×10^{-3} at a chosen time, in Navier-Stokes.

A distinction that the events of September 2026 make necessary. Finite-time blowup has now been claimed for the three-dimensional Navier-Stokes equations with a smooth force compactly supported in space and time [OAI26a], for the unforced Euler equations from smooth compactly supported data [OAI26b], and, in the same week, for forced Euler, Boussinesq and porous-medium flows [AB26], continuing the multiscale programme of Cordoba and Martinez-Zoroa [CMZ24]. A force of that kind satisfies Fefferman’s conditions (4) and (5), so a correct proof of the first would settle option (C) of the Clay problem [F00]; none of the claims had been independently verified as we wrote, and the accompanying formalizations are self-assessed. They do not speak to the object of this paper. Hou’s scenario is unforced, and the exclusion theorems we measure against [CSTY08, KNSS09, SS09] are theorems about the unforced equation, which no forced construction tests. One number transfers. The forced core of [OAI26a] is itself axisymmetric with swirl, and by our arithmetic from its stated scales it leaves the Koch-Nadirashvili-Seregin-Sverak hypothesis by a single power of the remaining time, with exponent below one part in a hundred; over our resolved range the same quantity does not move at all.

Two questions about these scenarios have not been asked in print. First, every stability statement about them, Hou’s data-perturbation tests, the nonlinear stability of Chen and Hou [CH25], and the unstable-direction counts of the machine-learned profiles of Wang and coauthors [WLGB23, W25], is a statement within the axisymmetric class; whether the collapsing state is stable to symmetry-breaking perturbations is unknown. Second, the viscous scenario’s relation to the exclusion theorems for Type I singularities of axisymmetric flows with swirl [CSTY09, KNSS09, SS09] has been argued about but not measured: Hou identifies a logarithmic growth of ru^r as the mechanism by which his solution escapes those theorems [H24], and Barker [B25] has responded that the theorems’ hypotheses do not involve that quantity.

This paper answers the first question and measures the quantities relevant to the second. Section 2 describes the codes and their validation. Section 3 gives the spectral table: the growth rates of azimuthal modes $m = 0, 1, 2$ about the collapsing state, in units of the collapse rate, with resolution certificates in both grid directions, together with the dependence of the $m = 1$ crossing on viscosity, the identification of the instability as the centrifugal instability of the front column, and a computer-assisted certificate of its discrete eigenvalue. Section 4 gives the nonlinear fate of the unstable perturbations. Section 5 gives the viscous continuation: the singular-time fit, the front-following map, and the identification of the front. Section 6 places the measured quantities against the theorems. Section 7 discusses what the results mean for constructions of Type II singularities and for the regularity side of the problem.

The results are negative in the sense the singularity literature cares about: the leading interior scenario is unstable to symmetry breaking, the instability depletes rather than intensifies the collapse, and the viscous continuation sits inside the class the theorems exclude. They are positive as characterization: the spectral table is the first non-axisymmetric stability computation of an axisymmetric blowup candidate, the front is identified as an exact local solution, and the theorem gap is stated with the measured constants attached.

2 Equations, codes and validation

2.1 Formulation

We use the Hou-Luo variables $u_1 = u_\theta/r$, $\omega_1 = \omega_\theta/r$, $\psi_1 = \psi/r$ on the periodic cylinder $r \leq 1$, $z \in [0, 1)$ with the axisymmetric Navier-Stokes equations in the form of [H23a, H23b], the same initial data (the swirl profile $12000(1 - r^2)^{18} \sin(2\pi z)/(1 + 12.5 \sin^2(\pi z))$), and the same boundary condition at the wall. The base solver (axiphys) integrates these equations on a radial sinh-type map and an axial map, with the Hou-Li exponential filter, a moving radial map that follows the collapse, and, for the runs of Section 5, a front-following axial map described there. It reproduces Hou’s published checkpoints: the Euler vorticity ratio at his first checkpoint to one percent and the Navier-Stokes ratio at $t = 0.0022616$ to three percent (x144 against his 139.6); the details of the cross-code comparison, including the spread among Hou’s own tabulated and fitted values, are in the supplementary record (Rounds 86-87).

2.2 Azimuthal-mode solvers

Perturbations of azimuthal wavenumber m about an axisymmetric base are computed in three ways. The frozen-base solver (axi3dlin) linearizes about a snapshot and integrates the mode with an exact discrete pressure projection (the divergence-of-gradient operator assembled from the same difference stencils, banded, with a Neumann wall row so that the projected radial velocity vanishes at the wall) and the parity rules $u'_r, u'_\theta \sim r^{m-1}$, $u'_z, p' \sim r^m$ at the axis; viscosity enters through the combinations $u_r \pm iu_\theta$, which obey the Laplacians of index $m \pm 1$. The tangent-linear solver (axi3dtl) marches the nonlinear axisymmetric base and the mode in one Runge-Kutta step, linearizing about the instantaneous base at every stage, with the base’s moving radial map installed as a fixed map and mapped bases transferred by exact sine series. The nonlinear solver (axi3dnl) carries modes $m = 1$ to $M - 1$ with the azimuthal convolutions evaluated pseudo-spectrally on $3M$ points and feeds the $m = 0$ Reynolds stress back into the base equations.

For $m = 0$ the seed exponents become $u'_r, u'_\theta \sim r$ and $u'_z \sim 1$, and the axial-mean Neumann problem for the pressure is singular and is pinned at the axis.

2.3 Gates

Solid-body rotation is neutral to 1×10^{-10} in the frozen solver. A Rayleigh-unstable column reproduces a one-dimensional primitive-variable eigensolver (0.3708 for $m = 0$; $0.248 + 0.337i$ for the physical $m = 1$ mode, with wall-localized spurious modes identified and excluded). The nonlinear code with a seed of 1×10^{-6} reproduces the tangent-linear rate to three digits (0.158 against 0.160) and generates the $m = 2$ harmonic at second order. The tangent-linear and frozen-base rates are compared on a common convention, the raw rate in inverse seconds at equal fraction of the remaining time: the two solvers normalize by different remaining times (the frozen one by the remaining time at the snapshot, the co-evolving one by the remaining time at the current time), so their printed columns are not comparable and a ratio taken between them is not a growth-rate ratio. On the common convention the co-evolving Euler $m = 1$ rate exceeds the frozen one by 4 percent at 2.2 percent of the remaining time and by 13 percent at 13 percent, and the excess then grows (Section 3.1); they agree to within 4 percent only over the first 6.5 percent of the remaining time, where the base has not yet moved, and by a fifth of it the co-evolving rate is 26 percent higher.

Amplification labels. Two different amplifications appear in the literature on this scenario and we use both, so we state which is which. States of the Euler and first-stage Navier-Stokes solutions are named by Hou’s tabulated amplification of the vorticity, which is how his checkpoints are identified,

so ‘the x144 state’ is his; quantities in Section 5 and after are named by the swirl amplification, the maximum of u_1 divided by its initial value, which is what our own continuation tracks. The two differ by more than an order of magnitude on the same snapshot, and mixing them was a defect of an earlier draft.

A standing rule, and the reason for it. Every quantity in this paper is reported with the range over which it was verified, and that convention was arrived at by repeatedly violating it. In the course of this work a rate was quoted beyond the interval where a resolution twin confirmed it; a length-ratio exponent was quoted before we noticed that three reasonable definitions of the two lengths give three different answers, two of them of opposite sign; a base state was measured on a sub-box holding a thousandth of the structure it was meant to characterize; and, in the certified computation of Section 3.6, three separate bounds that are adequate at two hundred basis functions proved inadequate at seven thousand, one of them by a factor that would have inverted its conclusion. The common failure is not arithmetic but scope: a quantity measured on part of an object, or at one size, reported as a property of the object. We therefore state, for every number below, the grid, the domain and the parameter range on which it was checked, and where a comparison requires us to choose a length scale we say so, because such a comparison is a choice and not a measurement.

2.4 The trivial directions and the threshold

On a Type I base $u = (T-t)^{-1/2}U(x/\sqrt{T-t})$ the amplitude rate σ of a perturbation, in units of the collapse rate $1/(T-t)$, is $3/2$ for the time-translation direction $d_t U$ and 1 for an axis translation; a perturbation outruns the collapse (its ratio to the base grows) when $\sigma(T-t) > 1/2$. That threshold is a floor the literature already guarantees, not a bar this paper is the first to clear: for a Type I Euler blowup Vasseur and Vishik’s Theorem 1 [VV20] bounds the maximal Lyapunov exponent of the base flow below by $1/2$ in exactly these units, and Lafleche, Vasseur and Vishik [LVV21] prove the axisymmetric version, that the exponent of the linearized flow diverges as t approaches T . Neither result identifies the perturbations that realize the floor, and neither bounds how far above it they lie. One caveat attaches to the comparison itself. The rate we measure is a global energy-norm rate for the perturbation, while the base is normalized pointwise, by the supremum of the velocity times the square root of the remaining time; the two are directly comparable only while the perturbation stays concentrated on the collapse scale. The diagnostics of Section 3.2 support that, the axial wavenumber centroid co-scaling with the collapse and the radial centroid sitting at the swirl maximum, but it is a modelling step and not an identity. What the computations below add is that identification – the long-wave displacement direction of Section 3.2, and the short-wave centrifugal family of the front of Sections 3.2 and 3.6 – together with the size of the excess, which is a factor of two to four at the last resolved point of every family we measured. For $m = 1$ the axis-translation field $-d_x U$ is an exact tangent-linear solution and is projected out with the energy inner product (exact for a normal operator; the overlap stays below 0.3 in all runs, so the projection error is at the ten percent level or below); for $m = 0$ the overlap with $d_t U$ is monitored and stays below 0.1. We report both the raw rate and the rate of the orthogonal part where it matters.

2.5 Certificates

Every long-wave entry of the table has three axial resolutions (512, 1024 and 2048 points, with the localized seed pinned at the same position) and a radial pair (an exact subsample of the sinh-map grid to 257 points against 513). The short-wave families have an axial pair (1024 against 2048) on the viscous base. The viscous continuation has a twin at a different axial map and Courant number, and the front-resolved runs have two maps. Three earlier readings were retracted during

the work and are recorded as such in the supplementary log: a fit-dependent ‘departure’ of the Type I constants (the singular-time degeneracy of Section 5.1), a grid-scale event in a seeded run at the x200 state, and a three-cell front thickness. We regard the retractions as part of the certificate.

3 The spectral table

3.1 Euler, the x28 state

The base is the Euler solution at $t = 0.002208$, where the swirl amplitude is 28 times its initial value, the core sits at $(R, Z) = (0.0250, 0.0078)$, and the Type I constant $\sup |u| (|x| + \sqrt{T - t})$ is 76 (it is 75 to 76 on every state from x28 to x200, set by the outer flow).

The convention on the Euler branch, stated because it is a convention. Collapse units in this section use $T_E = 0.002278$, Hou’s Euler checkpoint. That is not a fitted singular time and we do not present it as one. Running the log-derivative method of Section 5.1 on our densest Euler record (a 513×512 run with twenty-six retained snapshots to $t = 0.0022736$) does not converge: the last seven four- and five-point windows give T between 0.0022716 and 0.0022845 with the exponent drifting from 0.905 to 1.35, and neither stops moving as the window slides toward T , so Luo and Hou’s criterion is not met; a separate Euler run passes $t = 0.002278$ with an amplitude of 1.5×10^6 , still finite. We therefore do not invent an Euler fit. The consequence is a systematic band, not a correction: over that range of admissible T the frozen-base entries of this section move by about minus nine to plus nine percent and the late co-evolving ones by about minus seventeen to plus seventeen, and every Euler number below should be read with that band attached. What the Euler conclusions actually rest on is ratios—between resolutions, between azimuthal wavenumbers, between the frozen and the co-evolving base, and between the two-dimensional and the columnar rate—and every such ratio taken at equal time is invariant under the choice of T . Sections 3.2 to 3.6 are on the viscous branch and use the fitted $T = 0.0022865$ of Section 5.1 throughout.

Long-wave $m = 1$ (Figure 1 collects the Euler rates of this section; Figure 2 the resolution pairs). On the frozen base the localized seed grows at 0.184, 0.232, 0.275, 0.313, 0.346, 0.375, 0.401, 0.425, 0.448 at 2.2, 4.3, 6.5, 8.7, 10.9, 13.0, 15.2, 17.4, 19.5 percent of the remaining time; at three axial resolutions the rate at 8.7 percent reads 0.318, 0.313, 0.319 and at 13 percent 0.379, 0.375, 0.380 (512, 1024, 2048 points: agreement to 1.3 percent), and the radial subsample reads 0.306 at 8.7 and 0.351 at 11.6 percent (2 percent). On the co-evolving base the rate rises 0.25, 0.34, 0.42, 0.49, 0.56, 0.64, 0.71, 0.76, 0.82 at 5.7, 11.2, 16.6, 21.8, 26.9, 31.7, 36.4, 40.9, 46.3 percent. Set against the frozen rate on the common convention of Section 2.3—the raw rates at equal fraction, the printed columns being on different normalizations—the co-evolving base grows faster by 4 percent at 2.2 percent of the remaining time, 13 percent at 13.0, 38 percent at 23.9 and a factor 2.0 at 43.4, the last fraction at which both runs exist. The excess therefore grows through the run as the base moves away from the snapshot; it is not the constant thirteen percent that comparing the two printed columns suggests. After projecting out the translation direction (the solver’s energy projection of Section 2.4, re-measured from the logged overlap with a smoothed derivative) the orthogonal part reads 0.41, 0.48, 0.55, 0.63, 0.69, 0.75 at 23.1, 28.1, 32.9, 37.6, 42.0, 46.3 percent. The non-trivial $m = 1$ perturbation therefore outruns the Euler collapse from about 30 percent of the remaining time and reaches 0.75 of the collapse rate by 46 percent. Its axial wavenumber centroid co-scales with the collapse (7 to 30 cycles per unit as the core height shrinks), the swirl share of its energy production rises from 0.07 to 0.39, and its precession rate $\omega(T - t)$ passes from +0.12 through zero to retrograde near 40 percent; the swirling base has only $SO(2)$ symmetry, so this reversal is permitted and is not a bifurcation signature.

Long-wave $m = 2$. 0.31 at 17 percent on the frozen base, rising.

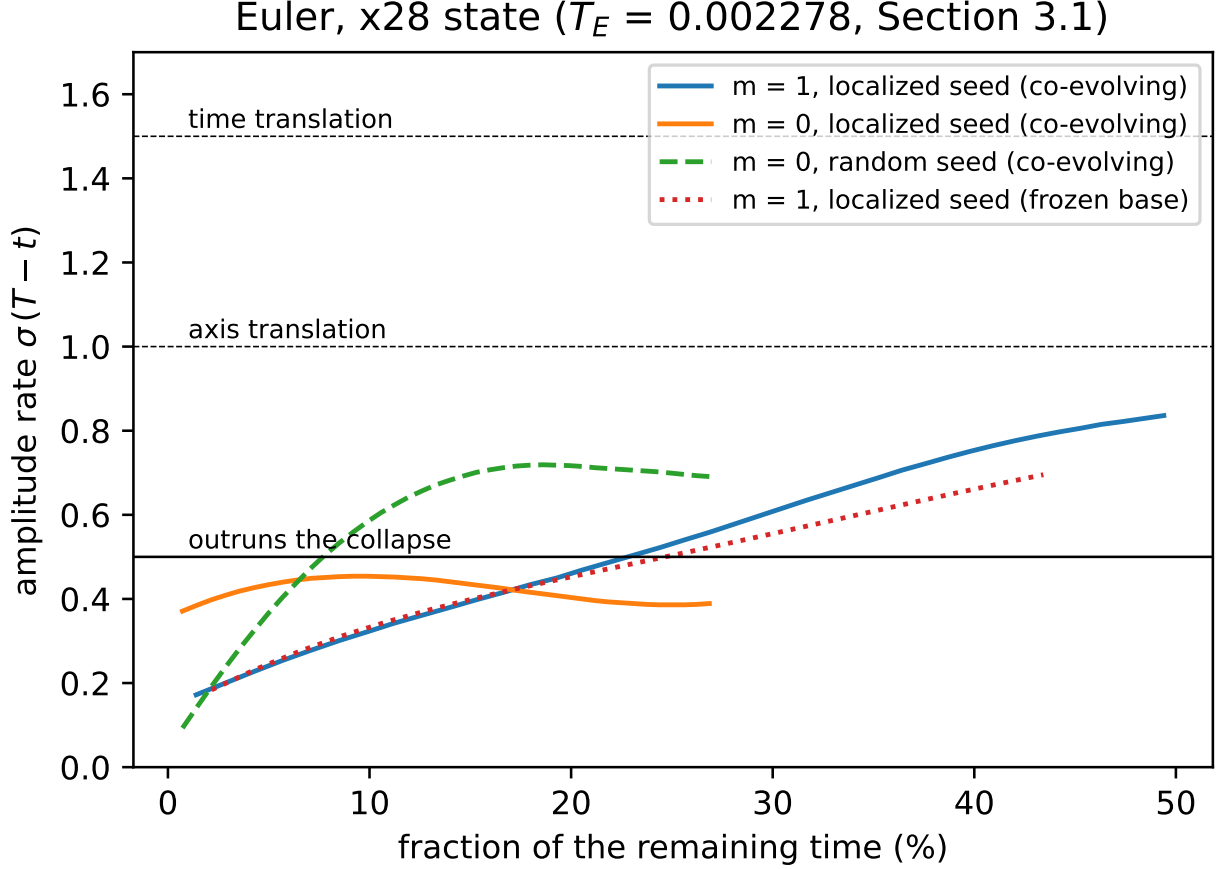


Figure 1: Euler, x28 state: amplitude rates $\sigma(T-t)$ of the $m=1$ and $m=0$ localized seeds and the $m=0$ random seed on the co-evolving base (tangent-linear), and of the $m=1$ localized seed on the frozen base, against the fraction of the remaining time. The line at $1/2$ is the threshold at which a perturbation outruns the collapse; the lines at 1 and $3/2$ are the axis-translation and time-translation directions.

Long-wave $m=0$. The localized seed on the frozen base rises to a maximum of 0.495 at 11.7 percent and falls to 0.378 by 29 percent; at three axial resolutions the maximum reads 0.495 , 0.496 , 0.497 (512, 1024, 2048 points) and the radial pair agrees to 1 percent (0.484 against 0.489 at 8.7 percent, 0.497 against 0.495 at 11.6). On the co-evolving base it plateaus at 0.454 at 10 percent and falls to 0.389 by 27 percent, with the overlap with $d_t U$ at 0.08 to 0.10 . The long-wave axisymmetric direction is stable: no non-trivial axisymmetric unstable direction exists at this level, consistent with Hou’s report that the profile is ‘very stable’ to perturbations of the data [H23b, Sec. 3.6].

Short-wave families. A smooth random seed under the same envelope excites, for every m , a family living in a thin band on the front whose axial wavenumber centroid marches with the collapse (17 to 80 cycles per unit for $m=0$; 26 to 54 for $m=1$ over the resolved range) and whose production is entirely from the meridional shear. Its rate has no maximum in Euler: for $m=1$ it reaches 0.9 of the collapse rate by 16 percent; for $m=0$ it crosses $1/2$ at 8 percent and reaches 0.72 by 17 to 20 percent before the axial resolution limits it. This is the Kelvin-Helmholtz continuum of the front, in Hou’s words the ‘shearing instability’ that ‘typically occurs near the sharp front’ and that his numerical viscosity is chosen to suppress [H23a, Sec. 3.6].

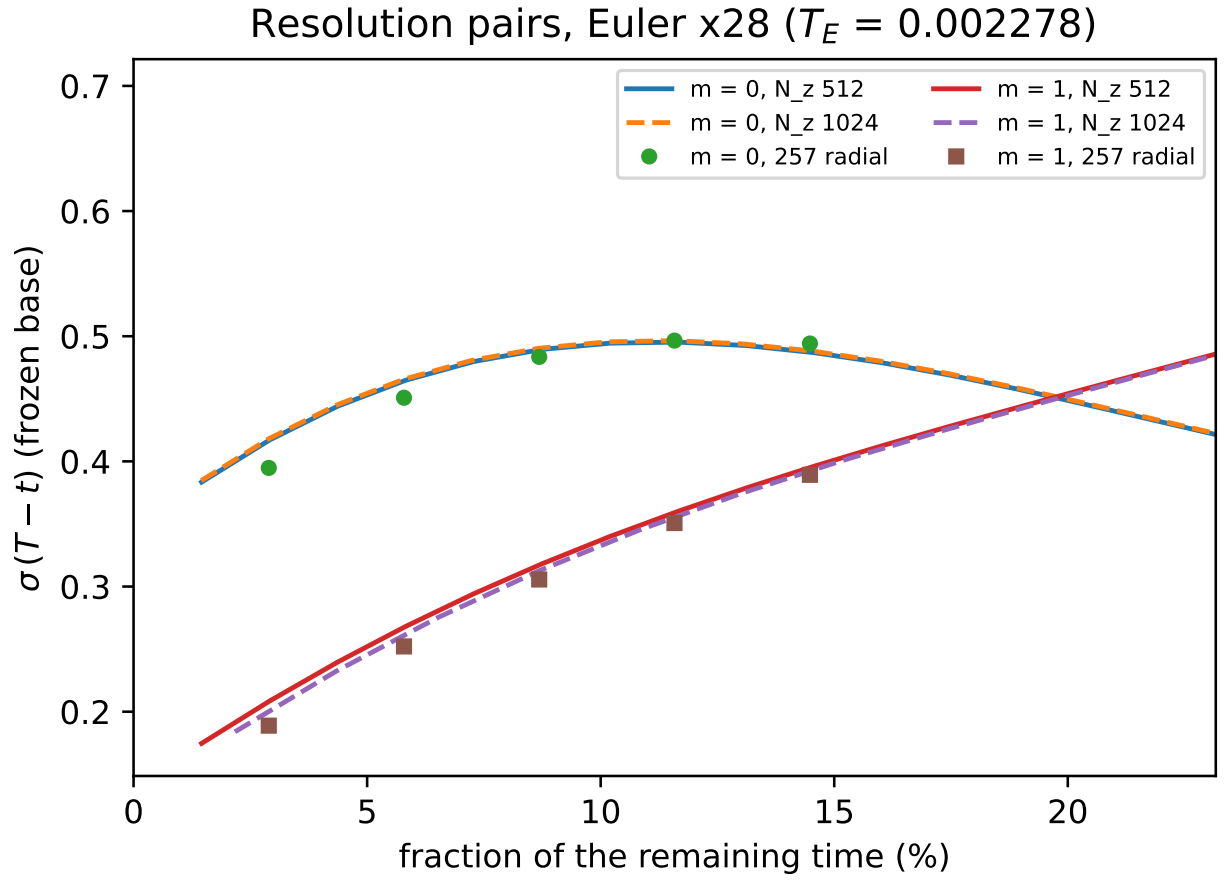


Figure 2: Resolution pairs on the Euler x28 state: frozen-base rates $\sigma(T-t)$ of the long-wave $m=0$ and $m=1$ entries at 512 and 1024 axial points and on the 257-point radial subsample.

3.2 Navier-Stokes, the x144 state, viscosity 5×10^{-4}

The base is Hou’s first-stage solution at $t = 0.002262$ (exactly $t = 0.0022620220$, amplitude 144 times the initial, his tabulated 139.6). Collapse units in this section and in Sections 3.3 to 3.6 use the fitted $T = 0.0022865$ of Section 5.1, so the remaining time on this state is $T - t = 2.4478 \times 10^{-5}$. The base amplitude grows locally like $(T - t)^{-1.3}$ there: the log-derivative over the pair of continuation points bracketing the state gives 1.27, the neighbouring pairs 1.09 and 1.35, and a power-law fit to the base amplitude across the whole tangent-linear window 1.38. The state is therefore Type I to about thirty percent in the exponent, and the converged $p = 1.03 \pm 0.03$ of Section 5.1 is reached only over the later, resolved range x131 to x331.

Long-wave $m = 1$ (co-evolving; Figure 3 collects the viscous rates of Sections 3.2 and 3.3): 0.19, 0.29, 0.38, 0.49, 0.60, 0.73, 0.86, 0.98, 1.18 at 1.0, 3.9, 6.7, 10.4, 14.0, 18.3, 22.5, 26.6, 34.2 percent; orthogonal part 0.36, 0.54, 0.77, 1.03 at 10.4, 18.3, 26.6 and 34.2 percent. Viscosity does not hold the long-wave mode below the threshold of Section 2.4 on this base. The rate crosses that threshold, half the collapse rate, at 10.7 percent of the remaining time, passes the collapse rate itself at 27.2 percent, and reaches 1.18, or 2.4 times the threshold, at 34.2; the translation-orthogonal part, which is the non-trivial content of the direction, crosses the threshold at 16.6 percent, passes one at 33.7 and ends at 1.03, or 2.1 times the threshold. That some direction clears the threshold is guaranteed in advance by [VV20, LVV21]; what is measured here is that the direction realizing it on the viscous base is an identifiable one, a displacement of the collapse, that it exceeds the guaranteed floor by a factor of two, and that it is still rising at the last resolved point. The long-wave direction on the viscous base is therefore unstable in the sense of Section 2.4, unlike the long-wave axisymmetric direction of Section 3.1.

Short-wave families (co-evolving, random seed): $m = 1$ reads 0.91, 1.17, 1.29 and 1.53 at 6.7, 9.5, 11.4 and 16.7 percent and 2.15 at 34.2, with the frozen-base axial pair agreeing to 3 to 7 percent (0.385 against 0.358 at 3.9 percent, 0.583 against 0.564 at 5.2); $m = 0$ crosses $1/2$ at 4.7 percent, the collapse rate at 15.2, and reads 1.48 at 23.9 percent; $m = 2$ crosses $1/2$ at 4.2 percent, the collapse rate at 6.9, and reads 2.08 at 23.9 percent. None of the three saturates within the resolved range: each is still rising at its last measured point, and the apparent flattening of the $m = 2$ curve under the earlier normalization was an artefact of dividing by a remaining time that shrank too fast. The front’s continuum is unstable relative to the collapse at every azimuthal wavenumber, the three crossing the threshold within half a percentage point of the remaining time of one another (4.2, 4.3 and 4.7 percent for $m = 2, 1$ and 0), although at equal fraction their rates then separate by a factor 1.3 to 1.7: it is a property of the sheet, not of a symmetry class. At the last resolved point the three stand at 1.48, 2.15 and 2.08 collapse units for $m = 0, 1$ and 2 , three to four times the threshold and all still rising, so the excess over the floor of [VV20, LVV21] is large and is carried by an identified family, the shear band on the front. The sheet Reynolds number $\Delta U / \sqrt{\nu a}$ with the measured velocity jump (Section 5) is 120 to 150 and, under the Type I scaling, constant in time, so the family is never frozen out.

3.3 Navier-Stokes, the x144 state, viscosity 5×10^{-3}

Hou raises the viscosity tenfold at $t_0 = 0.00227375$ because 5×10^{-4} ‘is not strong enough to stabilize the shearing instability generated by the 3D Euler equations’ [H23b, Sec. 3]. Applying 5×10^{-3} to base and mode from the x144 state, a random seed first decays (rates -2.0 in collapse units at the first output, 0.7 percent of the remaining time) and its surviving part re-grows: $m = 1$ crosses zero at 9.3 percent and $1/2$ at 14.5 percent, passes the collapse rate at 19.2 and reads 1.37 at 24.1 percent; $m = 0$ crosses zero at 12.4 percent and $1/2$ at 19.1 percent; the long-wave $m = 1$ direction

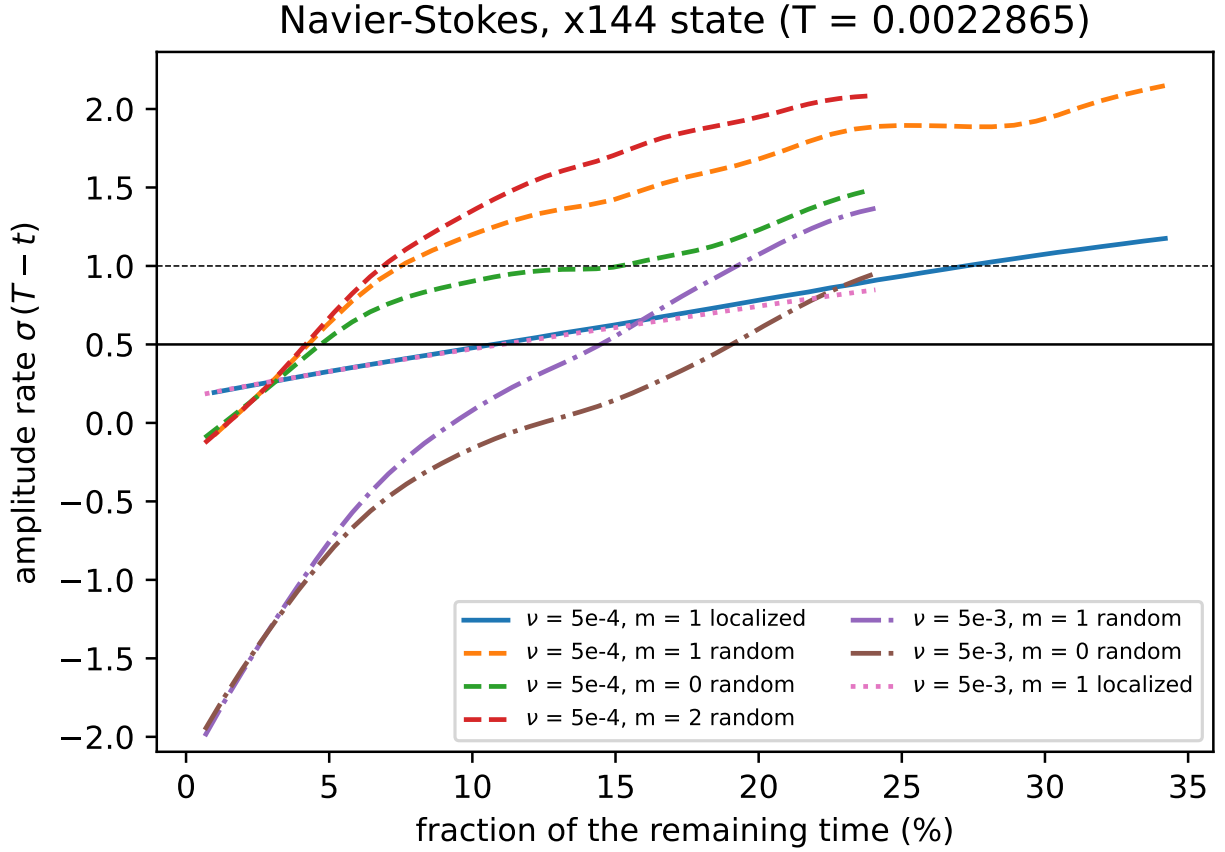


Figure 3: Navier-Stokes, x144 state: amplitude rates $\sigma(T-t)$ on the co-evolving base at $\nu = 5 \times 10^{-4}$ ($m = 1$ localized seed; $m = 0, 1, 2$ random seeds) and at $\nu = 5 \times 10^{-3}$ ($m = 0, 1$ random seeds; $m = 1$ localized seed), against the fraction of the remaining time. The line at $1/2$ is the threshold.

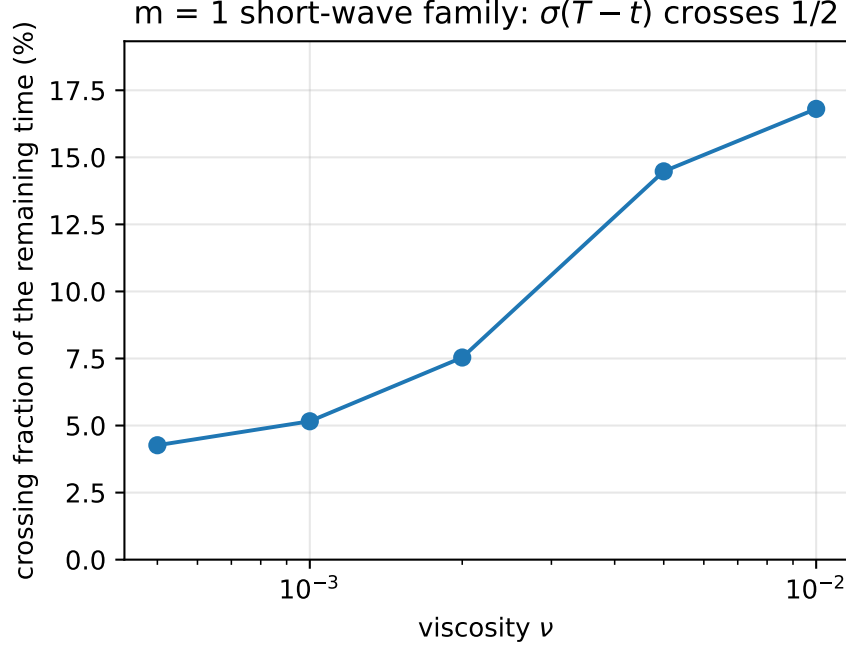


Figure 4: The $m = 1$ short-wave family on the co-evolving x144 base: the fraction of the remaining time at which $\sigma(T - t)$ crosses $1/2$, against the viscosity ν .

risks to 0.85 at 24.1 percent, crossing the threshold at 11.0 percent, so the tenfold viscosity does not hold that direction below the threshold either. The second-stage viscosity delays the axisymmetric family from 4.7 to 19.1 percent of the remaining time and the non-axisymmetric one from 4.3 to 14.5; it removes neither. At 24.1 percent the non-axisymmetric family stands at 1.37 and the long-wave direction at 0.85, or 2.7 and 1.7 times the threshold, so the tenfold viscosity thins the excess over the guaranteed floor without closing it. At equal fraction of the remaining time the base at 5×10^{-3} grows marginally more slowly than at 5×10^{-4} (amplitude 1.048 against 1.050 at 3.9 percent), and both are above the Type I line, which reads 1.041 there, consistent with the local exponent 1.3 measured in Section 3.2.

3.4 The crossing fraction versus viscosity

For the $m = 1$ random-seed family on the co-evolving x144 base, the fraction of the remaining time at which the rate crosses the threshold of Section 2.4, half the collapse rate (Figure 4), is 4.27 percent at 5×10^{-4} , 5.16 at 1×10^{-3} , 7.54 at 2×10^{-3} , 14.48 at 5×10^{-3} and 16.81 at 1×10^{-2} (the last run starting at -4.0 and crossing zero at 13.6 percent). Over the decade from 1×10^{-3} to 1×10^{-2} the crossing fraction rises by 11.7 points, roughly linearly in the logarithm of the viscosity and flattening at the top. At every viscosity from Hou’s first stage to twice his second, the non-axisymmetric family outruns the collapse before a fifth of the remaining time has elapsed. We do not extrapolate the trend to a viscosity that would hold the family below the threshold for half the remaining time: the five points span a factor twenty in viscosity and any such extrapolation lands far outside them, and an estimate of that kind carried by an earlier draft could not be reproduced from the data it was supposed to rest on.

3.5 An analytic cross-check

Leibovich and Stewartson’s sufficient condition [LS83] for non-axisymmetric short-wave instability of a swirling flow, $2V\Omega [D(rV) D\Omega + D(W^2)] < 0$ with V the swirl, $\Omega = V/r$, W the axial velocity and D the radial derivative, is evaluable on the axisymmetric base alone. It is satisfied in a thin annulus just outside the swirl maximum on every state examined: at $(r, z) = (0.0273, 0.0078)$ against the maximum at $(0.0250, 0.0078)$ on the Euler x28 state, at $(0.0141, 0.0037)$ against $(0.0138, 0.0037)$ on the viscous x144 state, and at $(0.0030, 0.0009)$ against $(0.0029, 0.0008)$ at x222; the implied short-wave growth is of order the collapse rate. Rayleigh’s centrifugal discriminant is negative on 6, 12 and 33 percent of the column through the swirl maximum inside twice its radius on the three states.

The evaluation at the viscous x144 state is made rigorous as a statement about an explicit field. Let $\Omega(s)$ and $W(s)$ be the degree-20 Chebyshev polynomials in $s = (r/r_{\max})^2$, $r_{\max} = 2R$, with exactly rational coefficients (the double-precision least-squares fit to the column through the swirl maximum, which reproduces the grid column to 1.9 percent in the swirl and 4.5 percent in the axial velocity), and let $(V, W) = (r\Omega, W)$ be the columnar vortex they define. Then $\Phi = 2r\Omega^2 B(r)$ with B a polynomial of degree 80 with rational coefficients, and by Sturm’s theorem in exact rational arithmetic B has no real root in $[0.013898, 0.015068]$ while B is negative at the midpoint; hence $\Phi < 0$ on $1.009R \leq r \leq 1.094R$. The explicit columnar field therefore satisfies Leibovich and Stewartson’s sufficient condition [LS83] for non-axisymmetric short-wave instability on that annulus. (The polynomial reproduces the column’s values to a few percent but its first derivative is wrong by up to a factor two at the swirl maximum, which matters for eigenvalues and not for this sign certificate; see Section 3.6. The computation is in the accompanying script; an interval-arithmetic evaluation of the Chebyshev series failed by the dependency problem at this degree, which is why the exact polynomial route is used. The same certificate on the Euler x28 column, degree 30 on $[0, 1.5R]$, holds on $0.998R \leq r \leq 1.197R$, but there the polynomial reproduces the grid column only to 16 percent in the swirl because of the sharper inviscid front; we report it as supplementary.)

3.6 The columnar model of the front and a certified eigenvalue

The columnar model. A column of the flow at fixed height, treated as a parallel swirling flow $(V, W) = (r\Omega, W)$ with axial velocity, has normal modes $\exp(im\theta + ikz - i\omega t)$ governed by a one-dimensional eigenvalue problem, which we solve by Chebyshev collocation with the axis conditions imposed by parity and no-slip at the outer edge of the fitted column ($2R$). Two cautions attach to the certificate of Section 3.5. The first concerns the field: the certificate is a statement about the explicit polynomial field; that field satisfies the criterion where the grid field does (the grid field’s discriminant is negative on the single band $1.00R$ to $1.81R$ with the same depth, the polynomial’s on wiggly sub-segments of it), but the polynomial’s first derivative is wrong by up to a factor two at the swirl maximum, and its normal-mode eigenvalues do not represent the flow’s: the columnar solver finds a fast near-front mode on the polynomial field and none on the grid column through the swirl maximum.

The front column. The second caution concerns the column: the one that carries the instability is the front’s, not the swirl maximum’s: on the grid column through the front at the height where the two-dimensional $m = 1$ mode peaks, the columnar solver has an isolated unstable viscous $m = 1$ eigenvalue with growth 1.24 in collapse units at $kR = 28$ (converged in resolution and domain, 20 percent above the next eigenvalue, inviscid limit 1.27 to 1.36 and still rising slowly with k), against 1.21 to 1.24 for the frozen-base two-dimensional rate at 13 percent of the remaining time. The columnar rate varies from 0.78 to 1.61 across the front’s thickness, the columnar eigenfunction

sits at 1.3 to $1.4R$ where the two-dimensional mode peaks at $1.0R$, and on the inviscid x28 state, whose front is sharper and whose instability is the axial-shear (Kelvin-Helmholtz) mode of the free front, no column reproduces the two-dimensional rate at all (columnar rates 0.4 to 0.7 against 0.92 , at twice the radius); so the parallel-flow model is local, identifies the mechanism on the viscous front and brackets rather than predicts the two-dimensional rate; but it is a discrete, well-separated eigenvalue of a one-dimensional problem, which is the object a rigorous enclosure can reach (Section 7.3).

Mechanism. Its mechanism is centrifugal: on the front column the circulation $r^2\Omega$ decreases outward beyond $1.2R$, so Rayleigh’s discriminant is negative there (on $1.21R$ to $2.0R$), and the inviscid $m = 1$ rate approaches the local centrifugal rate $\sqrt{-\Phi_R} = 1.47$ collapse units from below as k grows ($1.27, 1.34, 1.36$ at $kR = 22, 44, 88$) with the eigenfunction narrowing onto the radius of the bound’s maximum, which is the fixed- m large- k limit of Billant and Gallaire [BG05, BG13]; the viscous peak at $kR = 28$ is the Mayer-Powell cutoff of that branch [MP92]. That rate is the large-wavenumber centrifugal limit and not a universal bound: under strong axial shear the Leibovich–Stewartson rate exceeds it, so approaching it from below is a property of this column rather than a constraint on any column. The axial flow of the collapse extends the centrifugal instability to every azimuthal wavenumber [BG13], which is why the $m = 0, 1$ and 2 short-wave families of Section 3.2 cross the threshold within a few percent of each other (on the front column the viscous columnar peaks are $1.21, 1.24, 1.22$ and 1.16 collapse units for $m = 0, 1, 2, 3$, at $kR = 22, 28, 33$ and 44 (the $m = 3$ peak not yet passed), all in the same annulus: one centrifugal instability of the front at every azimuthal wavenumber, not three); without axial flow a Rayleigh-stable column is spectrally stable under further hypotheses [GS20], whose theorem asks for monotone vorticity besides the Rayleigh condition, adds a Richardson-type condition, and excludes the axisymmetric wavenumber, and no rigorous or computer-assisted result exists for an unstable eigenvalue of a swirling column with axial flow.

Figure 5 collects the columnar spectra of the front column against wavenumber at both viscosities and inviscid, with the Rayleigh bound, the two-dimensional rate and the certified points.

A certified discrete eigenvalue. We therefore certify the discrete columnar problem. Let $\Omega(s)$ and $W(s)$ be the degree-30 polynomials in $s = (r/r_{\max})^2$, $r_{\max} = 2R$, with exactly rational coefficients (the double-precision Chebyshev fit of the front column, converted exactly; the fit reproduces the grid column’s eigenvalue to 0.01 percent), and let (A_N, B_N) be the N -point Chebyshev-Gauss collocation pencil of the linearized viscous problem ($m = 1, k = 2000, \nu = 5 \times 10^{-4}$, no-slip imposed at the outermost collocation point $r_{\max} \cos(\pi/4N)$, axis conditions by parity) on the field $(r\Omega, W)$, with every entry enclosed in interval arithmetic. Then, by a Krawczyk verification with rigorous floating-point error bounds (Rump’s method; midpoint-radius matrices; all products bounded by the classical γ_n inequalities; the code and its tests accompany the paper), the pencil has a simple eigenvalue in the disc of radius 1.2×10^{-7} about $\omega = -3.5051747 \times 10^4 + 5.0850344 \times 10^4 i$ for $N = 150$, with its eigenvector in an enclosed box in which the eigenpair is unique (the disc is not certified to contain no other eigenvalue; the nearest floating-point eigenvalue is 1.05×10^4 away), and likewise for $N = 100$ (radius 5.2×10^{-8} , agreeing to nine digits) and $N = 60$ (5.0850368×10^4 , radius 2.4×10^{-8}). An independent assembly of the pencil at 200 bits without the parity trick agrees with every stored entry within its radius, and a separate symbolic check reproduces the Cartesian linearized equations on a test field to 1×10^{-34} . The assumptions are those of every interval computation on BLAS and mpmath: the real matrix product is a classical inner-product algorithm, the interval library is correct, overflow is detected, underflow is floored. Toward the continuous operator, the defect of the certified eigenvector decays spectrally ($2.4 \times 10^{-1}, 6.1 \times 10^{-3}, 5.2 \times 10^{-5}$ relative at $N = 60, 100, 150$) and on the divergence-free subspace the operator is nearly normal at this eigenvalue (inverse-norm constant 0.23 against 0.19 for a normal operator, in units

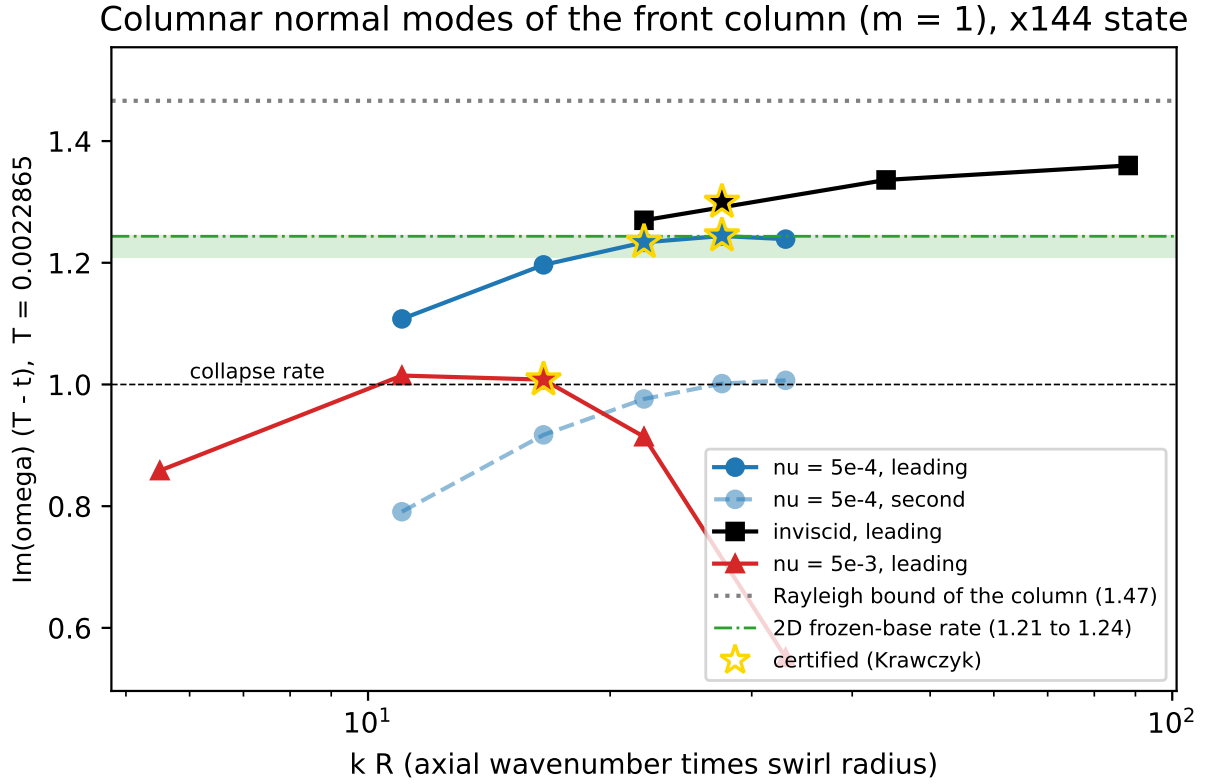


Figure 5: The columnar (parallel-flow) spectrum of the front column of the x144 viscous state versus axial wavenumber, in collapse units, at $\nu = 5 \times 10^{-4}$ (leading and second eigenvalue), inviscid, and $\nu = 5 \times 10^{-3}$; the Rayleigh bound of the column; the 2D frozen-base rate; the certified discrete eigenvalues (Krawczyk) at $k = 1600, 2000$ ($\nu = 5 \times 10^{-4}$), 1200 (5×10^{-3}), 2000 (inviscid).

of $1/(\nu k^2)$, identical at the three resolutions); what remains is the rigorous bound on that constant. The enclosure is a statement about the physical eigenvalue in inverse seconds and does not involve the singular time; only the conversion to collapse units does. In those units, with the fitted $T = 0.0022865$ and $T - t = 2.4478 \times 10^{-5}$ on this state, the certified growth rate is $1.244713 (T - t)^{-1}$ for each discrete problem. The same certificate at $N = 100$ gives 1.007974 at Hou’s second-stage viscosity 5×10^{-3} ($k = 1200$), 1.299945 for the inviscid problem ($k = 2000$), and 1.208120 for $m = 0$ ($k = 1600$, $\nu = 5 \times 10^{-4}$), each with radius below 1.3×10^{-7} , so the discrete columnar instability is certified at both viscosities, inviscid, and at two azimuthal wavenumbers. All four certified rates exceed the threshold of Section 2.4, half the collapse rate, by factors of 2.49, 2.02, 2.60 and 2.42, so what is certified on this column is not merely that the front is unstable, nor merely that some direction clears a floor that [VV20, LVV21] guarantee in advance, but that this identified direction, the centrifugal mode of the front column, clears that floor by a factor of two or more, at Hou’s first-stage viscosity, at his second, and inviscid. The margin is smallest at the second-stage viscosity 5×10^{-3} , where the certified rate 1.007974 is 2.02 times the threshold; that is the weakest of the four claims, and it is the one that bears on Hou’s continuation, since 5×10^{-3} is the viscosity he raises to. This is a computer-assisted proof about the discretized problem on an explicit field; it certifies neither the continuous columnar operator (that needs defect and inverse-norm bounds of the Nakao-Plum type, and the nine-digit agreement across N is evidence for it, not proof) nor the two-dimensional collapse, whose operator has no isolated eigenvalue to enclose (Section 3.7).

3.7 The spectrum of the frozen operator

The rates of Sections 3.1-3.4 are growth rates of evolved perturbations, not eigenvalues. A direct eigenvalue computation of the frozen $m = 1$ operator (Arnoldi iteration on the matrix-free linearization, largest real part) shows why the distinction matters. On the Euler x28 state at 257×512 the leading eigenvalues are complex pairs with $\sigma(T - t) = 3.14$ and 3.04 (frequencies ± 115 and ± 110 in collapse units), residual $\|Lv - \lambda v\|/\|v\| = 7 \times 10^{-6}$, with axial wavenumber centroid at half the resolved range and radial centroid just inside the swirl maximum: they are the short-wave shearing continuum of Section 7.1 at the grid scale, whose rate grows with resolution, and not the long-wave rate 0.3-0.5 that the localized seed excites. The long-wave direction is therefore not a leading discrete eigenvalue of the inviscid frozen problem, and an inviscid eigenvalue enclosure would be a statement about the grid.

The viscous operator ($\nu = 5 \times 10^{-4}$) damps the short waves and should have a resolution-convergent leading eigenvalue; that is the object a rigorous enclosure would target. Plain Arnoldi did not reach it: at 513×1024 the iteration failed to converge in 400 restarts, and at 257×512 with a Krylov space of 60 it had not returned after 8.5 CPU-hours (about 30,000 operator applications) and was stopped. The leading eigenvalues sit in a cluster whose separation is of the order of the viscous damping of neighbouring short waves, which is the regime in which unshifted Arnoldi stalls [MR96, LS01, T11]; the remedies are a shift-invert or Cayley transform with an inner linear solve, or the exponential (time-stepper) transform of Edwards, Tuckerman, Friesner and Sorensen [ETFS94], Arnoldi on $\exp(L\Delta)$, which needs no inner solve and contracts the damped continuum to the origin. We tried both matrix-free routes: Arnoldi on the propagator over 6.5 percent of the remaining time (190 applications, eight restart cycles) did not converge, and shift-invert with an inner GMRES solve stalled at a 36 percent inner residual for a real shift at the threshold ($\sigma(T - t) = 1/2$), with or without a shifted-diffusion preconditioner, which says the shifted operator is nearly singular over a region: the shift lies inside the pseudospectrum of the damped continuum rather than near an isolated eigenvalue. The leading viscous eigenvalue is therefore not isolated at this resolution, and this is the solver’s view of the same continuum that Section 7.1 describes; it also bounds what a

rigorous enclosure of the two-dimensional operator could achieve, since every enclosure radius scales with the inverse of the distance to that continuum. The reachable rigorous statement is the one-dimensional radial eigenvalue problem on the front column, certified at the discrete level in Section 3.6. The evolved-perturbation rates of Sections 3.2-3.3, certified by their resolution pairs, stand independently of this.

4 Nonlinear fate

The $m = 1$ mode was seeded at 20 and 50 percent of the local swirl velocity on the Euler x28 state, at 20 percent on the viscous x144 state, and at 20 percent on the Euler x200 state, with $m = 0, 1, 2$ (and 3) carried and the Reynolds-stress feedback on the base. Against the unperturbed collapse at matched times the axisymmetric amplitude lags: on Euler x28 by 0.8, 1.3, 2.1, 3.6 percent at 10, 13, 16, 21 percent of the remaining time with the 20 percent seed and by 5 percent at 11 percent with the 50 percent seed, at which the axisymmetric swirl maximum stalls and falls while the three-dimensional swirl maximum stays flat at 1.9 times the axisymmetric one, the mode's own growth drops below its linear rate (0.26 against 0.31), and the $m = 2$ harmonic sits on the ϵ^2 curve at 8×10^{-3} of the $m = 1$ energy without migrating. The x28 continuation reached 49 percent of the remaining time with a 22 percent lag and the three-dimensional swirl ratio flat at 1.37. On the viscous base, by contrast, the feedback produces no measurable lag: at every output time the two runs share, out to 34 percent of the remaining time, the seeded run's axisymmetric amplitude agrees with the unperturbed collapse to better than 0.03 percent, so the depletion measured here is a property of the Euler states and not of the viscous one within the range we resolve.

A random seed on the viscous x144 base (three modes, seed 0.2, followed to 23.9 percent of the remaining time) tests whether the short-wave family rolls up into an intensifying substructure of the kind seen in strained mixing layers and tornado subvortices. The three-dimensional vorticity peak rose from 1.04 to 1.71 times the axisymmetric peak while the axisymmetric amplitude ran 9 percent above the Type I line and the second harmonic stayed at two percent of the first. The end-state fields show the mode's energy density peaking on the ring (radius 0.98 of the swirl maximum's, height 0.91 of its) with half-widths of 0.7 to 0.8 collapse lengths, thirty times the Burgers thickness $\sqrt{\nu(T-t)}$, and the swirl at that point exceeding its azimuthal mean by a factor 1.38. No compact off-axis maximum at the viscous scale, no Burgers-locked core, no climbing harmonics: the rising vorticity ratio is the linear family growing on a base whose amplitude it has not yet measurably touched.

The outcome is the generic saturating Reynolds-stress feedback of a supercritical mode (Stuart-Watson [SW60]), the mechanism by which Hou and Li's simulation of Kerr's antiparallel tubes [HL06, HL08] was depleted. A perturbation of relative size ϵ reaches order one after the base amplifies by $\epsilon^{-1/\lambda}$ with $\lambda = \sigma - 1/2$. On the corrected normalization λ is 0.34 for the long-wave $m = 1$ direction of the Euler x28 state, 0.68 for the long-wave direction of the viscous x144 state and 1.65 for its short-wave $m = 1$ family, so a one percent seed reaches order one after an amplification of about 9×10^5 on the Euler long-wave direction, 9×10^2 on the viscous long-wave one and 16 on the viscous short-wave family. On the viscous base that is well inside the range a computation resolves, and it is what the runs show: the mode reaches order one relative to the base swirl rather than the collapse outrunning it, although at the last point we reach the base amplitude itself is not yet measurably depleted. We found no seed, state or viscosity at which the symmetry-broken state continued to collapse.

5 The viscous continuation

5.1 The singular time and the amplitude exponent

Hou’s own caveat [H23b, Sec. 3.4] is that the Type I rate inequalities ‘would be almost impossible to verify numerically since it requires the exact value of T ’. We fit T by the log-derivative method: for $A \sim C(T - t)^{-p}$ the ratio A/A' equals $(T - t)/p$, linear in t with slope $-1/p$ and zero at T , so sliding-window fits give p and T together with nothing assumed, and Luo and Hou’s criterion [LH14a, LH14b] is that both stop moving as the window slides toward T . On the resolved snapshots of the continuation (Section 5.2) four-point windows ending at 0.002281, 0.002282, 0.002283, 0.002284 and 0.0022845 give $p = 1.49, 1.37, 1.19, 1.04, 1.05$ with $T = 0.0022888, 0.0022881, 0.0022872, 0.0022865, 0.0022866$, and five-point windows ending at 0.0022845, 0.002285 and 0.0022853 give $p = 1.04, 1.00, 1.06$ with $T = 0.0022865, 0.0022864, 0.0022866$: both have stopped moving over the last three windows. The exponent is not constant along the way, and measuring it locally resolves what would otherwise look like a disagreement with Section 3.2. Centred differences of $\log A$ against $\log(1/(T - t))$ on the fourteen recorded states give a local exponent that rises from 1.07 at 3 times the initial swirl amplitude to a maximum of 1.33 near 25 times, which is why the state used in Section 3 carries 1.3, and then falls monotonically through 1.12, 1.03 and 0.99 to 0.98 by 92 times. A fit over all fourteen returns 1.20 and a fit over the last six returns 1.007. So the amplitude approaches the Type I exponent from ABOVE, with a transient overshoot of about a third in the middle of the range, and the value quoted here is the settled one, at $p = 1.03 \pm 0.03$ and $T = 0.0022865 \pm 1 \times 10^{-7}$. The earlier apparent $p \sim 1.8$ of a fixed- T fit was the concave A/A' of subleading corrections in the early windows. A first reading of ‘falling Type I constants’ from a $1/A$ fit with a drifting T was retracted on this basis.

5.2 The continuation and the front-following map

Hou’s first-stage solution was continued at $\nu = 5 \times 10^{-4}$ past his switch time on a 513×1024 grid with a moving radial map and an axial map concentrated at the symmetry plane, from 25 to 217 times the initial amplitude, with a twin at a different axial map and Courant number agreeing to 0.5 percent in amplitude through x217; past x217 the front’s width fell to two cells and the twins diverged by 27 percent, and the run was stopped. A 2048-point restart from x131 with a three-times finer axial map read the front at three cells at x268, still grid-limited. The solver was then given a front-following axial map: a Gaussian point density centred on the vorticity maximum with a fifth of the points within three front widths, re-fitted with the radial rezone every hundred steps, the fields moved by exact sine-series evaluation; a map contrast above about 500 breaks the mapped Poisson solve, so the window is kept at three front widths (contrast about 140, symmetrization error 2×10^{-5}), at which the restart reproduces the state to a percent.

On that map the same x222 state has a front 4.8×10^{-5} wide over ten cells where the coarse map had read 7.7×10^{-5} over three, and after one interval (x268) the front is 3.3×10^{-5} wide over nine cells. Re-evaluating the x268 state on a second map gives the error bars: front thickness ± 15 percent, vorticity plateau ± 5 percent, swirl constant ± 2 percent from the grid, meridional constant ± 20 percent, amplitude and its constants 0.5 percent.

With the window floor set at 1.5×10^{-4} (contrast 140 to 170) the continuation was carried on two grids (1536 and 2048 axial points) from 131 to 395 times the initial amplitude, agreeing to 0.5 percent, which is the range of every measurement in Sections 5.3 to 6. At about 400 times the initial amplitude ($T - t$ about 8×10^{-7} , $\log(1/(T - t)) = 14.0$) the one-scale collapse ends: the primary front stalls (x395 to x382 over one output interval, $A(T - t)$ falling from 4.3 to 3.7 with $T = 0.0022866$) while a second structure nearer the symmetry plane, at z about 0.0003 against the

front's 0.0009, grows to 96 percent of it; at the next interval the maximum is at the second structure (x484, at $r = 0.00185$, $z = 0.00030$), and at the next it has burst to x776 ($r = 0.00158$, $z = 0.00025$, $A(T-t) = 5.6$). The two follower-map grids and a third run on a wide fixed fine region (2048 points, no following, contrast 175, which cannot lose the front) agree at every one of these points to 0.2 to 0.6 percent through x484 and to 3 percent at x776, so the transition is a property of the solution and not of the map. The aspect ratio z/r of the maximum's position drops from 0.42 to 0.16 across it, the swirl inside the maximum empties ($|u_1|$ at 0.7 of the maximum's radius, on its row, falls from 15 percent of the maximum at x395 to 3 percent at x776, and at 1.05 of the radius from 83 to 28 percent, so the structure is a ring compact in r), and the meridional velocity at the maximum turns from axial to radially inward ($|u^r|/|u^z|$ from 0.05 to 20 to 67), which is the opening of the vacuum region between the front and the axis that Hou's Sec. 3.2.4 describes his data as avoiding. This has the sign of Hou's two-scale structure: in his Case 4 at the same viscosity (different data) a vacuum region for u_1 opens between the front and the axis and the growth then slows under viscous dominance [H23b, Secs. 1.3, 3.2.4 and 3.6], and in his rescaled model at dimension three the one-scale profile drifts toward two scales, which he calls a mild scaling instability [H24, Secs. 1 and 5]; whether a vacuum region opens here, and whether u^r/u^z at the maximum follows the aspect ratio, is what the wide-map run's later points will show. Past the transition the collapse does not resume. The wide-map run and the two follower runs agree to one percent at every point through 909 times the initial amplitude, and the amplitude sequence 395, 382, 484, 776, 899, 909, 1025, 1422, 2501 carries $A(T-t) = 4.3, 3.7, 4.1, 5.6, 5.4, 4.4, 3.7, 3.4, 3.0$ with $T = 0.0022866$: the constant of Section 5.3 is replaced by pulses of about 60 percent, local fits of the exponent give a singular time bouncing between 0.0022861 and 0.0022895 over consecutive intervals, and the front and the ring alternate as the site of the maximum. No exponent is determined after 400 times the initial amplitude, and those last points are on maps past their tested contrast. Followed further on a wide fixed map with ten times denser output, the state does not arrest. The amplitude rises through 1390, 1764, 2546, 2861, 3991 and 5079 times the initial swirl, monotonically over the first fourteen of the sixteen recorded points, the vorticity maximum with it, and the structure descends onto the symmetry plane and contracts. The run passes the singular time fitted before the transition, with local fits of the logarithmic derivative giving a consistent later time of 0.0022867; the last two points fall back to 4337 and 4763, so the pulsing of the post-transition state persists and the growth at the stopping point is not monotone. A twin at half the axial resolution instead peaks and decays ninefold, and a twin at twice the radial resolution reaches a third again higher amplitude at the same instant, so the decay is the coarse grid suppressing the collapse and not the collapse ending. We stop there, at between four and six thousand times the initial amplitude, because the structure by then is resolved by seven cells across its width in one direction or the other on every grid we can run, against the ten-cell standard used above, and because the one cheap remedy, a tighter axial window, would take the map contrast past the value at which the mapped Poisson solve is known to fail. Two scale-invariant diagnostics are unchanged throughout that range: the circulation maximum stays pinned near 64.5 by its maximum principle, and the local L^3 norm of the velocity, which the theorem of Escauriaza, Seregin and Sverak [ESS03] requires to become unbounded at a singularity, stays flat at 47.1 while the amplitude grows by a factor three. The measurements of Sections 5.3 to 6 are all made before the transition, and the range over which the one-scale law and the constants hold is 144 to 395 times the initial amplitude.

5.3 The constants

With $T = 0.0022865$ the resolved snapshots from x131 to x331 give $A(T-t) = 3.85, 3.86, 4.00, 4.06, 4.10, 4.15, 4.16, 4.14$ (flat); $R/\sqrt{T-t} = 2.38, 2.40, 2.34, 2.34, 2.33, 2.36, 2.29, 2.20$; $\sup |u|\sqrt{T-t}$

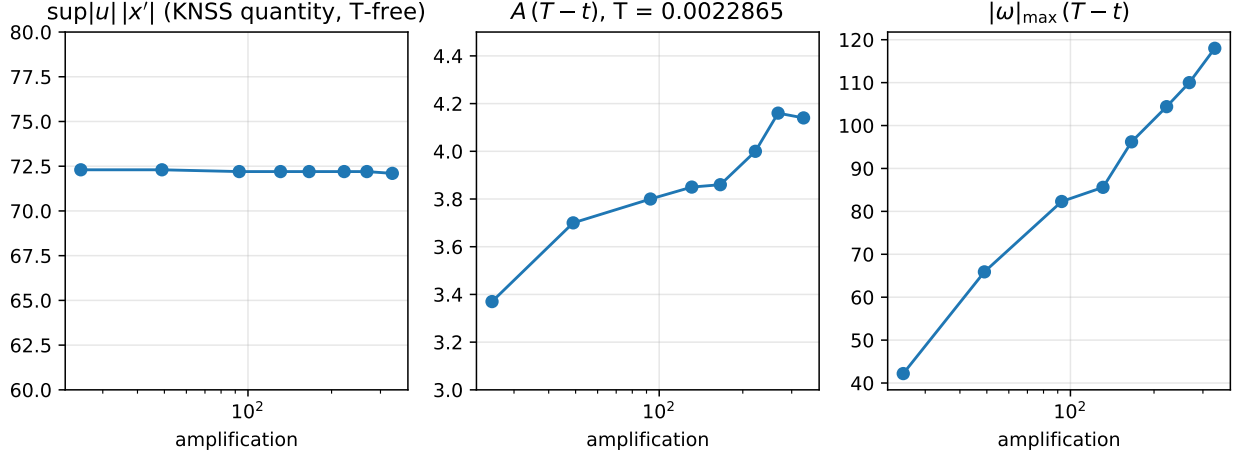


Figure 6: The constants against amplification (recorded values): $\sup|u||x'|$ (the KNSS quantity, T -free), $A(T-t)$ with $T = 0.0022865$, and $|\omega|_{\max}(T-t)$.

between 9.2 and 10.4, of which ± 2 percent is grid and the rest the singular-time sensitivity (a 1×10^{-7} change in T moves it 2 percent at $T-t = 2 \times 10^{-6}$ and more later), so that its growth law is not readable from this data to the precision that would distinguish a logarithmic modulation from a small power or from a T error; the meridional Type I constant $\sup|(u_r, u_z)|\sqrt{T-t} = 6.0$ to 7.7 (bounded); $\omega(T-t) = 82, 86, 96, 104, 106, 110, 118$ (the x331 value from the run with the window floor; a second run whose window tightened without a floor degraded numerically at that point and is discarded), with its growth exponent in $1/(T-t)$ falling from 0.43 (x131 to x222) to 0.33 (x268 to x331): levelling toward its Burgers ceiling; $\omega/A = 21.4, 23.7, 25.5, 26.5, 28.5$ (T -free); the circulation maximum 64.9 to 64.7 (the maximum principle).

5.4 The front is the Rott-Lundgren layer

The front's full width at half maximum is 0.86, 0.65, 0.62, 0.39, 0.29 collapse lengths $\sqrt{T-t}$ and 3.9, 2.9, 2.8, 1.7, 1.3 Burgers lengths $\sqrt{\nu(T-t)}$ from x131 to x268 (the first three grid-limited), and 1.0 at x331 (six cells). The compressive strain at the front, measured on the symmetry plane at the ring radius and on the axis, is 2.2 to 3.8 in units of $1/(T-t)$. A Gaussian fit of the azimuthal vorticity across the front at x268 (24 cells inside three standard deviations) gives a standard deviation of $0.58\sqrt{\nu(T-t)}$, a relative residual of 5 percent and a baseline of 9 percent of the peak; the strain implied by the thickness, $\nu(T-t)$ over the variance, is 2.9 against the measured 2.2 to 3.3; the velocity jump implied by the fit, $\sqrt{2\pi} \delta \omega_{\text{peak}} = 2600$, agrees with the measured jump of u_r across the layer (2980) and with the integral of the vorticity (3240). The jump constant $\Delta U \sqrt{T-t}$ is 2.8 to 3.3, a third of the swirl constant, and the sheet Reynolds number $\Delta U / \sqrt{\nu a}$ is 120 to 150. At x331 (six cells) the fit residual is 10 percent and the implied strain 5.5 against a measured 3.8, and the u_r jump no longer accounts for the vorticity integral as the sheet tilts (the ratio of the radial to the azimuthal vorticity maximum rising from 1.1 to 2.1): the identification holds to the resolution available.

The front is therefore the exact strained-layer solution $u = (-ax, v(x, t), az)$, $v_t - axv_x = \nu v_{xx}$, of Rott and Lundgren [R58, L82], under the collapse strain, with thickness $\sqrt{\nu/a}$. This is the configuration in which the peak vorticity scales exactly as $1/(T-t)$ for a Type I velocity jump, and the vorticity's super- $1/(T-t)$ growth between x25 and x268 was its approach to this floor.

5.5 The local L^3 norm

The scale-invariant quantity, the L^3 norm of the velocity in the ball of radius four collapse lengths about the axis point the ring converges to, was computed on the resolved snapshots from x4 to x268: 25.3, 25.0, 24.6, 24.1, 23.3, 23.2, 23.1, 23.0, 22.8, 22.6, 22.5, 22.4, 22.3, 22.0, a monotone decrease of 13 percent over four units of $\log(1/(T-t))$, with the balls of eight and sixteen collapse lengths decreasing faster. What this does and does not say is stated in Section 6.3.

6 The scenario against the theorems

6.0 The T -free measurement that decides the class

The Liouville theorem of Koch, Nadirashvili, Seregin and Sverak [KNSS09] for bounded ancient axisymmetric solutions with swirl (their Theorem 5.3) has one hypothesis beyond boundedness: $|v(x, t)| \leq C/|x'|$ for all $t < 0$, with $|x'|$ the distance to the axis. If $\sup_t \sup_x |u(x, t)| |x'|$ is finite up to T , the Leray rescalings about any axis point obey $|u^{(\lambda)}(y, s)| \leq K/|y'|$ uniformly, so a Type I limit is a bounded ancient axisymmetric solution satisfying the hypothesis and is trivial, contradicting the normalization of the rescaling; and a point that is not Type I is handled directly by Theorem 6.1 of the same paper and by [CSTY08], which need no Type I assumption. Either way the scenario is regular; no bound on ru^r enters. The quantity to measure is therefore $\sup |u| |x'|$ over the domain as a function of amplification, and unlike every velocity constant of Section 5.3 it does not involve the fitted singular time at all. On the resolved snapshots it reads 72.3, 72.4, 72.2, 72.3, 72.3, 72.3, 72.2, 72.2, 72.2, 72.2, 72.2, 72.1 over 3.3 units of $\log(1/(T-t))$, from 9 to 331 times the initial swirl amplitude, constant to 0.42 percent while $\log(1/(T-t))$ rises from 10.6 to 13.9; the value at the swirl maximum itself is 21 to 22; the Type I constant $\sup |u| (|x| + \sqrt{T-t})$ is 75.4 to 74.6; and the circulation maximum is 64.9 to 64.7 by its maximum principle. Figure 6 plots these constants over the wider range the continuation covers, from 25 to 331 times the initial swirl amplitude, and shows a subset of the tabulated values. That last number is the cleanest contrast with the forced construction of [OAI26a], whose exterior swirl is a radial heat solution decaying like the radius to the power $-1-2h$, so that the circulation at its core diverges like the remaining time to the power $-h$; ours cannot grow at all, and does not. Three things must be said about what this does and does not show. First, the logical status: the theorems (Chen, Strain, Tsai and Yau [CSTY08]; Koch, Nadirashvili, Seregin and Sverak [KNSS09], Theorem 6.1; Seregin and Sverak [SS09]) exclude a singularity if $\sup |u| |x'|$ stays bounded up to T (the companion note states the proposition and its proof by citation, including the step from the periodic cylinder to the local theorems, which we flag for the analysts); a finite computation shows the bound over the resolved range, here 3.3 units of $\log(1/(T-t))$, and can establish membership in the excluded class only in that sense: what it shows is the absence, over that range, of the departure the scenario needs. Second, where the value comes from: the supremum 72 is attained in the outer flow (the circulation maximum sits at r about 0.40, z about 0.10, far from the core), while the same quantity at the swirl maximum is 21 to 22; so the bound the theorem needs is carried by the outer flow, and a logarithmic growth of the core quantity would not reach the supremum until it had grown by a factor of about three. The core quantity itself, $|u| r$ at the swirl maximum, reads 21.1, 22.1, 21.8 at 9, 130 and 266 times the initial swirl amplitude (20.5 at 331, where the map limits it), flat within its scatter, where a logarithmic growth over the range would read about 30 percent. Third, what Hou's own diagnostics do: the log-weighted circulation criteria of Lei and Zhang, $\max |\log r|^{3/2} |\Gamma|$ and $\max (\log r)^2 |\Gamma|$ over $r < 0.1$, grow steadily in our run (163 to 223 and 314 to 486 over the earlier part of the continuation), as they must for a core of constant circulation collapsing toward the axis, since the weight itself grows;

they are not measurements of the hypothesis of any exclusion theorem. The quantity Hou names as the escape is the global supremum of $|ru^r|$ in his generalized rescaled model [H24, Sec. 1.5 and 5.4.4], reported to grow like $\log(1/(T-t))$ over amplifications from 1×10^{15} to 1×10^{30} ; his three-dimensional Navier-Stokes paper [H23b] reports no value of it, and our range of amplifications has no counterpart in his data, so the comparison is structurally unavailable here. For the record, $|ru^r|$ restricted to three swirl radii about the maximum is non-monotone over our range: 21.9 at the start, falling to 14.9 at 72 times the initial amplitude, rising to 21.7 at 266 times (a 45 percent rise over 1.3 units of $\log(1/(T-t))$, faster than logarithmic but from its minimum) and 19.1 at 329 times on the other map, with the two maps differing by 13 percent at the end; $|ru^z|$ at the core rises slowly from 7.3 to 10.3; the global maxima (49 and 20) sit in the far field. A later pair of continuations at 1536 and 2048 axial points on the same map gives 15.2 to 18.0 over 263 to 395 times, the two grids agreeing to 3 percent, before the front-following map loses the front (the collapse region carries two maxima in z that a single-centre map cannot hold, which is what ended every continuation past about 330 times). Across the late-stage transition of Section 5.2 (x382 to x776, 0.34 units of $\log(1/(T-t))$, on the wide-map run) the core-restricted $|ru^r|$ rises from 16.5 to 20.7 within three swirl radii and from 25.6 to 32.3 within $r < 0.1$, $|ru^z|$ at the core from 9.8 to 16.0, and $|u|r$ at the swirl maximum from 20.9 to 24.1, while the global supremum $\sup |u||x'|$ stays in the far field; so Hou’s quantity moves in his direction at the regime change, by a quarter in a third of a log-unit, which is a jump between regimes and not yet a trend. On this quantity our data are therefore inconclusive: they show neither a sustained logarithmic growth nor its exclusion, and a comparison with Hou’s observation would need a continuation many orders of magnitude longer on one map with the core window fixed in collapse units. The computed scenario therefore satisfies, as measured over the resolved range and independently of T , the hypothesis of the exclusion theorems, with the bound carried by the outer flow and the core value flat; whether the core quantities eventually grow is the one thing the data leave open.

6.1 The hypotheses of the exclusion theorems

Chen, Strain, Tsai and Yau [CSTY09, II] prove that axisymmetric strong solutions with swirl in $\mathbb{R}^3$ satisfying $|v| \leq C_*|t|^{-1/2}$, with C_* arbitrary, are regular; no bound of the form $|u| \leq C/r$ and no bound on ru^r appears among the hypotheses. Seregin and Sverak [SS09], Theorem 1.1, prove the local statement: for an axisymmetric weak solution in L_3 with pressure in $L_{3/2}$, a Type I bound on the meridional velocity alone on a parabolic cylinder about an axis point gives regularity there; the hypothesis $|v| \leq C/|x'|$ appears only in their Theorem 3.2; and their Lemma 3.3 places singular points on the axis. Hou’s gloss that the theorems apply ‘provided that $|ru^r|$ and $|ru^z|$ remain bounded’ [H24, Sec. 1.5], and his identification of the logarithmic growth of $|ru^r|$ and of the log-weighted circulation as the scenario’s escape, therefore do not touch the hypotheses that are actually used; Barker [B25] makes the same observation in $\mathbb{R}^3$ and flags the wall at $r = 1$ as the point to check, which the locality of the Seregin-Sverak theorem [SS09] settles. The measured meridional constant of Section 5.3 is bounded. The escape the theorems leave is growth of $\sup |u||x'|$ itself, or of the meridional Type I constant, not of ru^r (Section 6.0); for a velocity that is not asymptotically Type I the Liouville theorem for bounded ancient axisymmetric solutions with swirl is open [KNSS09, AB19] but Theorem 6.1 of [KNSS09] and [CSTY08] still apply under the $|x'|$ bound; the sharpest log-supercritical criterion with swirl, Seregin’s [S21], allows growth only like $(\ln \ln 1/(T-t))^{1/224}$. Our data cannot read the swirl constant’s growth law to that precision (Section 5.3), so this escape is neither established nor excluded; every resolved observable, the amplitude exponent, the meridional constant, the front at the Burgers scale, is that of a Type I collapse.

6.2 The exact local solution and where finite energy acts

The layer of Section 5.4 is an exact Navier-Stokes solution for any strain history $a(t)$ (Rott 1958 [R58]; the Lundgren transformation [L82] reduces it to the heat equation), and for $a = 1/(T - t)$ its thickness is $\sqrt{\nu(T - t)}$ exactly. It belongs to the stretched-vortex class of Gibbon, Fokas and Doering [GFD99], in which finite-time singularities exist (Ohkitani-Gibbon; Constantin’s infinite family of exact Euler solutions of infinite energy [C00]; Maekawa, Miura and Prange’s backward self-similar Burgers-vortex blowup for Navier-Stokes with linear strain [MMP19]). These escape every theorem by having infinite energy: the KNSS Liouville theorem [KNSS09] needs bounded ancient solutions, and the layer rescales to a linearly growing one; the Seregin-Sverak argument [SS09] needs the rescaled limit to be a suitable weak solution; the Leray-profile theorems of Necas, Ruzicka and Sverak [NRS96] and of Tsai [T98] need a profile in L^3 or with local energy control. Finite energy acts at a computable distance: a linear strain $1/(T - t)$ gives velocity $L/(T - t)$ at distance L , while a Type I bound limits it to $M/\sqrt{T - t}$, so the linear profile survives only to $L \sim M\sqrt{T - t}$, about four ring radii, inside the computational domain. The depletion is visible in the numbers: the free exact layer has circulation growing like $T/(T - t)$ and a Type II velocity, ours has a Type I velocity and $p = 1$, so the layer’s circulation exponent is halved by the finite-energy feedback. A core-resolving computation can never falsify the local model, which is exact; it can falsify the global one, and what it shows is a Type I collapse with the front at the viscous scale and no departure to 331 times the initial amplitude.

6.3 What the local L^3 norm does and does not say

The decrease of Section 5.5 contradicts no theorem as stated. The global conditions (a singularity requires the whole-space L^3 norm to blow up [ESS03, S12]) say nothing about shrinking balls; Barker and Prange’s logarithmic lower bound [BP21] concerns balls of radius $(T - t)^{(1-\delta)/2}$ with a doubly exponentially small constant; the at-scale theorems (Barker-Prange 2020 [BP20], Theorem 2; Kang, Miura and Tsai [KMT21]) are lower bounds by a small universal constant, and the measured 22, which is 4.4×10^4 in the dimensionless units of the theorems, satisfies them by a wide margin. A Type I profile predicts the scaled norm to approach a constant, and a mild decrease toward a plateau is consistent with that. The sentence the data supports is that the critical norm’s content at the collapse scale is not growing, the opposite of the trend a singularity drives into the global critical norm: evidence against, not proof.

7 Discussion

7.1 The continuum obstruction

Every rigorous construction of a log-corrected (Type II) singularity in a parabolic or dispersive problem follows one recipe: a stationary or self-similar profile of a limiting problem, its linearized spectrum with finitely many unstable directions and a resonant mode, and a modulation argument. For Navier-Stokes the limiting problem is Euler, viscosity being exactly marginal under the Leray scaling, and this framing exists only as an exclusion program (Seregin’s Type II papers, Barker and Prange [BP20, BP21]), never as a construction. Our table supplies the missing spectral ingredient for the leading profile and finds that it fails: the inviscid profile has no non-trivial long-wave unstable direction but a continuum of unstable short-wave directions at every azimuthal wavenumber, whose in-frame rate passes the threshold of Section 2.4, at which a perturbation outruns the collapse, from 8 percent of the remaining time on the Euler state and from 4 percent on the viscous one, and exceeds it by a factor of three to four by the last resolved point (the threshold is itself the floor

[VV20, LVV21] guarantee; what the table adds is that the directions realizing it are these identified short-wave families of the front, and how far above the floor they run). This is the essential spectrum of Lifschitz and Hameiri [LH91], Friedlander and Vishik [FV91], and Vishik [V96] (whose radius equals the maximal Lyapunov exponent of the profile flow), which Chen and Hou reconcile with their stable boundary profile by working in a low-regularity weighted norm where the outgoing transport damps short waves, declining the high-regularity estimate whose coercivity they call unclear. A construction on the interior profile must either find such a norm or cut the continuum off with a viscosity large enough, which is what Hou’s tenfold viscosity does by hand only in part: by Section 3.3 it delays the axisymmetric family from 4.7 to 19.1 percent of the remaining time and the non-axisymmetric one from 4.3 to 14.5, and by Section 3.4 no viscosity we tested, up to twice his second stage, removes either. The object that results is a viscously selected profile whose inviscid limit is not an attractor, and an exactly self-similar viscous profile is what the Leray-profile theorems forbid. A postscript, from work posted while this paper was being written. The same continuum is the ENGINE of the constructions of [OAI26a] and [OAI26b], which state that they amplify short waves on a background flow by the criteria of Lifschitz and Hameiri and of Friedlander and Vishik, and design the core to supply enough shear at its outer edge to amplify pulses placed there. In [OAI26b] the mechanism runs unforced, on Euler, by iterating the amplification on successively shorter intervals so that the shear produced by one oscillation amplifies the next; the result is a gradient blowup with bounded velocity, not a self-similar profile. So the family we identify as the obstruction to a self-similar unforced scenario is, used the other way, a route to a non-self-similar one. Our measurements bear on that route more directly than we first supposed. An iteration of this kind compounds against the collapse exactly when the rate in the units of Section 3, the growth rate times the remaining time, exceeds the threshold of Section 2.4, one half, and its margin is the excess over that. Every wavenumber we measured on the co-evolving viscous state clears the threshold inside the resolved range, and none of the three does so marginally: q reaches 1.48 for $m = 0$, 2.15 for $m = 1$ and 2.08 for $m = 2$ at the last point of each run, all three still rising, and each is above the threshold from 4.7, 4.3 and 4.2 percent of the remaining time respectively and above the collapse rate itself from 15.2, 7.5 and 6.9 percent. The margins $q - 1/2$ at the last measured point are 0.98, 1.65 and 1.58 for $m = 0, 1$ and 2 . At the $m = 1$ margin a one percent seed needs 2.8 e-folds of the logarithm of the reciprocal remaining time to reach order one, an amplification of about 16; at the $m = 2$ margin, 2.9 e-folds and about 18; and even the slowest of the three, $m = 0$, needs 4.7 e-folds and an amplification of about 110. Over the 3.3 units of $\log(1/(T - t))$ that our continuation resolves, a mode at the $m = 1$ margin gains a factor 230 on the collapse, one at the $m = 2$ margin 180, and one at the $m = 0$ margin 25. The mechanism their construction uses is therefore not merely present in the unforced interior scenario: on this state it compounds easily, and it does so at every azimuthal wavenumber rather than at one. That is consistent with our nonlinear runs rather than in tension with them. A mode that reaches order one within three e-folds is a mode that arrives while the collapse is still resolvable, and what the nonlinear runs then find is that it depletes the collapse rather than riding it (Section 4). What the measurement does not supply is the construction. Compounding against the collapse is necessary for an iteration of the kind [OAI26b] performs and not sufficient for it: their iteration requires the shear amplified at one stage to be handed to the next, whereas here the amplified mode spends itself on the base, which is exactly the difference between a forced or engineered background and a collapse that must sustain itself.

7.2 The Liouville problem and the log escape

Whether a conditional Liouville theorem could close the log escape was examined directly. The KNSS argument [KNSS09] uses $|v| \leq C/|x'|$ at two lines, both as a statement uniform over rescalings, and a logarithmic weakening breaks both structurally for every exponent, not by a constant; the published growth-side results reach only a double logarithm to a small power (Pan’s exponent 0.028 [P16]; Seregin’s $1/224$ in scaled norms [S21]), and the counterexample of Seregin, Silvestre, Sverak and Zlatos [SSSZ12] for drift-diffusion Liouville theorems sits at the next rung. The one hypothesis type that would enter the proof at the level of the source term, a one-sided monotonicity of Γ^2 in the axial direction, is not satisfied by the computed profile: on the collapse side of the core the sign of $d_z(\Gamma^2)$ is mixed (58 to 69 percent negative by weight). The escape reduces to a single unextracted number, the dependence of the axis Holder exponent in the critical-drift oscillation decay on the drift constant; polynomial dependence would close every logarithmic power below a threshold, exponential dependence none. We leave it as the one analytical question this scenario poses.

7.3 What remains open

Discretely self-similar and aperiodic (chaotic) Type I singularities are excluded only near the continuous case (Chae and Wolf [CW17a, CW17b]) and are equivalent, in the aperiodic case, to the Liouville problem for bounded ancient solutions, which is open in three dimensions; but a discretely self-similar singularity is Type I with a point singularity, and in the axisymmetric swirl class every Type I singularity is excluded, so this loophole lives in the non-axisymmetric class, where the relevant computation, a period-log(λ) or aperiodic orbit of the rescaled equations in three dimensions, has never been attempted for a blowup problem (the closest precedent is Gundlach’s computation of the Choptuik spacetime as a periodic orbit in similarity time [G95]). The octahedral Kida-Pelz class is unexcluded by any theorem but its Euler collapse was superseded by depletion at higher resolution, and no viscous arrest study of it exists.

On the rigorous side of our own results, Section 3.6 certifies the discrete columnar eigenvalue; the step from the discrete to the continuous columnar operator is the Nakao-Plum programme (a defect bound for the certified eigenfunction and a rigorous bound on the inverse of the shifted operator on its complement), carried out for Orr-Sommerfeld and Rayleigh-Benard problems at this size; that programme has since been carried out here but for one computation and one argument, both stated below, and will be reported separately: the Stokes spectrum of the mode is certified complete out to fifty times the eigenvalue by an argument-principle count of 29278 zeros, the Galerkin matrix of the perturbation is enclosed rigorously in the first 7682 Stokes modes, the bordered inverse is certified at that size, the truncation beyond the block is controlled by an identity of Rellich type that makes the pressure coefficients explicit, and the acceptance constant is 0.04 against a limit of 1 when the eigenvector’s defect is taken at its computed floating-point value. Two things are outstanding, and we state them because the second is not merely a computation: the Galerkin strips between the block and the truncation, which is a finite calculation of known cost now running, and a proven bound on that defect. The bounds available for the latter control a whole strip rather than the single vector, which costs eight orders of magnitude and leaves the criterion unmet as proved, although the defect measured on the certified part of the strip is of order 1×10^{-6} , five orders inside the requirement. Closing that gap needs a single-vector argument and two further quadratures. Its completion would make the result the first rigorous unstable eigenvalue of a swirling column with axial flow. Neither step certifies the two-dimensional collapse, whose operator has no isolated eigenvalue to enclose (Section 3.7).

7.4 The regularity side

No finite list of computations certifies global regularity for all data, but certification for one datum is a finite computation in principle (Chernyshenko, Constantin, Robinson and Titi [CCRT07]; Dashti and Robinson [DR08]), discharged in practice only for mild data; for Hou’s datum the exponential tolerance collapses under a 10^7 amplification. What this work contributes to that side is characterization: a finite-codimension picture in which the collapse is Type I with a converged singular time, the front sits at the viscous scale, the axisymmetric long-wave direction is stable, and the short-wave families, axisymmetric and not, outrun the collapse at both of Hou’s viscosities and deplete it when they arrive.

7.5 Predictions for a three-dimensional computation

No non-axisymmetric computation of Hou’s data exists. The first one will test the following statements, which are the content of this paper reduced to numbers that a single run at $\nu = 5 \times 10^{-4}$, started from the axisymmetric solution at 144 times the initial amplitude with a small random perturbation, can confirm or refute.

1. The azimuthal wavenumber $m = 1$ energy in a thin band on the front grows, in the collapsing frame, faster than the collapse from 4.3 percent of the remaining time ($\sigma(T - t) = 0.91$ at 6.7 percent and 1.29 at 11.4 percent against the threshold $1/2$), with $m = 0$ and $m = 2$ following at 4.7 and 4.2 percent, and all three exceed the collapse rate itself, at 15.2, 7.5 and 6.9 percent. The long-wave $m = 1$ direction (a displacement of the collapse) also crosses the threshold, at 10.7 percent, and reaches 1.18, or 2.4 times the threshold, by 34 percent. The threshold is $1/2$ here as everywhere, and clearing it is guaranteed by [VV20, LVV21]; what is to be tested is the factor of three to four by which these families exceed it, and which wavenumbers carry them.
2. The growth stays a distributed deformation of the ring, not a sub-vortex: the three-dimensional vorticity maximum reaches 1.7 times the axisymmetric one while the axisymmetric amplitude stays on its own law, 9 percent above the Type I line (20 percent seed at 23.9 percent of the remaining time); the second harmonic stays at about two percent of the first; the mode’s energy peaks on the ring at 0.98 of the swirl maximum’s radius and 0.9 of its height with half-widths of 0.7 to 0.8 collapse lengths; no compact off-axis maximum forms at the viscous scale.
3. For a seed of relative size ϵ the symmetry breaking becomes order one after an amplification of $\epsilon^{-1/\lambda}$ with $\lambda = \sigma - 1/2$ between 0.68 and 1.65 on the viscous state, so a 1 percent seed reaches order one after an amplification between about 16 and 9×10^2 of the amplitude at which it was introduced; the collapse does not continue through it.
4. At Hou’s second-stage viscosity 5×10^{-3} the same instability crosses the threshold at 14.5 percent of the remaining time and the collapse rate at 19.2 (1.37, or 2.7 times the threshold, at 24 percent); the tenfold viscosity moves the crossing, it does not remove it.
5. Any resolved axisymmetric continuation of the $\nu = 5 \times 10^{-4}$ solution gives $\sup_x |u| |x'| = 72 \pm 1$ with no trend over 3.3 units of $\log(1/(T - t))$, the circulation maximum 64.7 to 64.9, the meridional Type I constant 6 to 8, an amplitude exponent 1.03 ± 0.03 against a converged singular time $0.0022865 \pm 1 \times 10^{-7}$, a front of thickness 1.0 to $1.7\sqrt{\nu(T - t)}$ fitting the Rott-Lundgren profile [R58, L82] to 5 percent, and $\omega(T - t)$ rising through 118 with a decreasing exponent.

6. On one map with the core window fixed in collapse units, the core-restricted $|ru^r|$ (within three swirl radii of the maximum) either stays within its present band of 15 to 22 or grows like $\log(1/(T - t))$ beyond 331 times the initial amplitude; a resolved continuation to 1000 times adds 1.1 units of $\log(1/(T - t))$ and decides between the two at the 10 percent level, which the present data cannot. (Partly tested after the prediction was written: over 382 to 776 times, across the late-stage transition of Section 5.2, the quantity rose from 16.5 to 20.7, above the band, in 0.34 log-units; whether that continues is the open part.)

A three-dimensional run in which the symmetry-broken state continues to collapse with its amplitude on the Type I law, or an axisymmetric continuation in which $\sup |u| |x'|$ grows logarithmically (about thirty percent over the range above), would refute the corresponding statement and with it the paper's reading of the scenario.

Data and code availability

All solvers, logs, snapshots at the reported times, the extraction scripts for the figures, and the dated record of results, corrections and retractions are in the repository accompanying this paper.

References

- [AB19] D. Albritton, T. Barker, On local Type I singularities of the Navier-Stokes equations and Liouville theorems, *J. Math. Fluid Mech.* 21 (2019), art. 43. [arXiv:1811.00502]
- [B25] T. Barker, Quantitative classification of potential Navier-Stokes singularities beyond the blow-up time, arXiv:2510.20757 (2025).
- [BP20] T. Barker, C. Prange, Localized smoothing for the Navier-Stokes equations and concentration of critical norms near singularities, *Arch. Ration. Mech. Anal.* 236 (2020), 1487-1541. [arXiv:1812.09115]
- [BP21] T. Barker, C. Prange, Quantitative regularity for the Navier-Stokes equations via spatial concentration, *Comm. Math. Phys.* 385 (2021), 717-792. [arXiv:2003.06717]
- [BG05] P. Billant, F. Gallaire, Generalized Rayleigh criterion for non-axisymmetric centrifugal instabilities, *J. Fluid Mech.* 542 (2005), 365-379.
- [BG13] P. Billant, F. Gallaire, A unified criterion for the centrifugal instabilities of vortices and swirling jets, *J. Fluid Mech.* 734 (2013), 5-35.
- [BK96] K. N. Beronov, S. Kida, Linear two-dimensional stability of a Burgers vortex layer, *Phys. Fluids* 8 (1996), 1024-1035.
- [CH22] J. Chen, T. Y. Hou, On stability and instability of $C^{1,\alpha}$ singular solutions to the 3D Euler and 2D Boussinesq equations, *Comm. Math. Phys.* 405 (2024), art. 112. [arXiv:2206.01296]
- [CH25] J. Chen, T. Y. Hou, Singularity formation in 3D Euler equations with smooth initial data and boundary, *Proc. Natl. Acad. Sci. USA* 122 (27) (2025), e2500940122; Stable nearly self-similar blowup of the 2D Boussinesq and 3D Euler equations with smooth data I: analysis, arXiv:2210.07191; II: rigorous numerics, *Multiscale Model. Simul.* (2025), doi 10.1137/23M1580395. [arXiv:2305.05660]

- [CW17a] D. Chae, J. Wolf, Removing discretely self-similar singularities for the 3D Navier-Stokes equations, *Comm. Partial Differential Equations* 42 (2017), 1359-1374. [arXiv:1610.09464]
- [CW17b] D. Chae, J. Wolf, On the Liouville type theorems for self-similar solutions to the Navier-Stokes equations, *Arch. Ration. Mech. Anal.* 225 (2017), 549-572. [arXiv:1609.06962]
- [CCRT07] S. I. Chernyshenko, P. Constantin, J. C. Robinson, E. S. Titi, A posteriori regularity of the three-dimensional Navier-Stokes equations from numerical computations, *J. Math. Phys.* 48 (2007), 065204. [arXiv:math/0607181]
- [C00] P. Constantin, The Euler equations and nonlocal conservative Riccati equations, *Int. Math. Res. Not.* 2000 (9), 455-465. [arXiv:nlin/0004019]
- [AB26] L. Alpoge, T. Buckmaster, Finite-time blowup for forced Euler, Boussinesq and incompressible porous media, manuscripts at cims.nyu.edu/~tristanb, 8 September 2026 (with H. Coiculescu for the porous-media case; not on arXiv).
- [CMZ24] D. Cordoba, L. Martinez-Zoroa, W. Zheng, Finite time singularities to the 3D incompressible Euler equations for solutions in C^∞ , arXiv:2407.06776.
- [F00] C. L. Fefferman, Existence and smoothness of the Navier-Stokes equation, Clay Mathematics Institute Millennium Prize Problem description, 2000.
- [OAI26a] OpenAI, Finite time blowup for Navier-Stokes, 165 pp., openai.com/index/navier-stokes-solution, 8 September 2026, with Lean formalizations at github.com/openai/NavierStokesAndEuler.
- [OAI26b] OpenAI, Finite time blowup for the Euler equation, 45 pp., openai.com/index/navier-stokes-solution, 8 September 2026, with Lean formalizations at github.com/openai/NavierStokesAndEuler.
- [CSTY08] C.-C. Chen, R. M. Strain, H.-T. Yau, T.-P. Tsai, Lower bounds on the blow-up rate of the axisymmetric Navier-Stokes equations, *Int. Math. Res. Not.* 2008, art. rnn016.
- [CSTY09] C.-C. Chen, R. M. Strain, T.-P. Tsai, H.-T. Yau, Lower bounds on the blow-up rate of the axisymmetric Navier-Stokes equations II, *Comm. Partial Differential Equations* 34 (2009), 203-232. [arXiv:0709.4230]
- [DR08] M. Dashti, J. C. Robinson, An a posteriori condition on the numerical approximations of the Navier-Stokes equations for the existence of a strong solution, *SIAM J. Numer. Anal.* 46 (2008), 3136-3150.
- [E21] T. M. Elgindi, Finite-time singularity formation for $C^{1,\alpha}$ solutions to the incompressible Euler equations on $\mathbb{R}^3$, *Ann. of Math.* 194 (2021), 647-727. [arXiv:1904.04795]
- [ESS03] L. Escauriaza, G. Seregin, V. Sverak, $L_{3,\infty}$ -solutions of the Navier-Stokes equations and backward uniqueness, *Russian Math. Surveys* 58 (2003), 211-250.
- [ETFS94] W. S. Edwards, L. S. Tuckerman, R. A. Friesner, D. C. Sorensen, Krylov methods for the incompressible Navier-Stokes equations, *J. Comput. Phys.* 110 (1994), 82-102.
- [FV91] S. Friedlander, M. M. Vishik, Instability criteria for the flow of an inviscid incompressible fluid, *Phys. Rev. Lett.* 66 (1991), 2204-2206.

- [GS20] T. Gallay, D. Smets, Spectral stability of inviscid columnar vortices, *Anal. PDE* 13 (2020), 1777-1832.
- [GFD99] J. D. Gibbon, A. S. Fokas, C. R. Doering, Dynamically stretched vortices as solutions of the 3D Navier-Stokes equations, *Physica D* 132 (1999), 497-510.
- [GHDG08] T. Grafke, H. Homann, J. Dreher, R. Grauer, Numerical simulations of possible finite time singularities in the incompressible Euler equations: comparison of numerical methods, *Physica D* 237 (2008), 1932-1936. [arXiv:0711.2284]
- [G95] C. Gundlach, The Choptuik spacetime as an eigenvalue problem, *Phys. Rev. Lett.* 75 (1995), 3214-3217. [arXiv:gr-qc/9507054]
- [H23a] T. Y. Hou, Potential singularity of the 3D Euler equations in the interior domain, *Found. Comput. Math.* 23 (2023), 2203-2249. [arXiv:2107.05870]
- [H23b] T. Y. Hou, Potentially singular behavior of the 3D Navier-Stokes equations, *Found. Comput. Math.* 23 (2023), 2251-2299. [arXiv:2107.06509]
- [H24] T. Y. Hou, Nearly self-similar blowup of generalized axisymmetric Navier-Stokes and Boussinesq equations, arXiv:2405.10916 (2024).
- [HL06] T. Y. Hou, R. Li, Dynamic depletion of vortex stretching and non-blowup of the 3-D incompressible Euler equations, *J. Nonlinear Sci.* 16 (2006), 639-664.
- [HL08] T. Y. Hou, R. Li, Blowup or no blowup? The interplay between theory and numerics, *Physica D* 237 (2008), 1937-1944.
- [KMT21] K. Kang, H. Miura, T.-P. Tsai, An epsilon-regularity criterion and estimates of the regular set for Navier-Stokes flows in terms of initial data, *Pure Appl. Anal.* 3 (2021), 567-594. [arXiv:2006.13145]
- [KNSS09] G. Koch, N. Nadirashvili, G. Seregin, V. Sverak, Liouville theorems for the Navier-Stokes equations and applications, *Acta Math.* 203 (2009), 83-105. [arXiv:0709.3599]
- [LS01] R. B. Lehoucq, A. G. Salinger, Large-scale eigenvalue calculations for stability analysis of steady flows on massively parallel computers, *Int. J. Numer. Methods Fluids* 36 (2001), 309-327.
- [LS83] S. Leibovich, K. Stewartson, A sufficient condition for the instability of columnar vortices, *J. Fluid Mech.* 126 (1983), 335-356.
- [LH91] A. Lifschitz, E. Hameiri, Local stability conditions in fluid dynamics, *Phys. Fluids A* 3 (1991), 2644-2651.
- [LVV21] L. Lafleche, A. F. Vasseur, M. Vishik, Instability for axisymmetric blow-up solutions to incompressible Euler equations, *J. Math. Pures Appl.* 155 (2021), 140-154. [arXiv:2009.12603]
- [LZZ17] Z. Lei, Q. S. Zhang, N. Zhao, Improved Liouville theorems for axially symmetric Navier-Stokes equations, *Sci. Sin. Math.* 47 (2017), 1183-1198 (Chinese edition; English text arXiv:1701.00868).

- [L82] T. S. Lundgren, Strained spiral vortex model for turbulent fine structure, *Phys. Fluids* 25 (1982), 2193-2203.
- [LH14a] G. Luo, T. Y. Hou, Potentially singular solutions of the 3D axisymmetric Euler equations, *Proc. Natl. Acad. Sci. USA* 111 (2014), 12968-12973.
- [LH14b] G. Luo, T. Y. Hou, Toward the finite-time blowup of the 3D axisymmetric Euler equations: a numerical investigation, *Multiscale Model. Simul.* 12 (2014), 1722-1776.
- [MMP19] Y. Maekawa, H. Miura, C. Prange, On stability of blow-up solutions of the Burgers vortex type for the Navier-Stokes equations with a linear strain, *J. Math. Fluid Mech.* 21 (2019), art. 46. [arXiv:1807.10341]
- [NRS96] J. Necas, M. Ruzicka, V. Sverak, On Leray's self-similar solutions of the Navier-Stokes equations, *Acta Math.* 176 (1996), 283-294.
- [MR96] K. Meerbergen, D. Roose, Matrix transformations for computing rightmost eigenvalues of large sparse non-symmetric eigenvalue problems, *IMA J. Numer. Anal.* 16 (1996), 297-346.
- [MP92] E. W. Mayer, K. G. Powell, Viscous and inviscid instabilities of a trailing vortex, *J. Fluid Mech.* 245 (1992), 91-114.
- [N84] J. C. Neu, The dynamics of stretched vortices, *J. Fluid Mech.* 143 (1984), 253-276.
- [P16] X. Pan, Regularity of solutions to axisymmetric Navier-Stokes equations with a slightly supercritical condition, *J. Differential Equations* 260 (2016), 8485-8529. [arXiv:1410.6260]
- [PV26] B. Pineau, V. Vicol, On rotated backwards self-similar solutions of the incompressible 3D Navier-Stokes equations, arXiv:2607.09619 (2026).
- [R58] N. Rott, On the viscous core of a line vortex, *Z. Angew. Math. Phys.* 9 (1958), 543-553.
- [S12] G. Seregin, A certain necessary condition of potential blow up for Navier-Stokes equations, *Comm. Math. Phys.* 312 (2012), 833-845. [arXiv:1104.3615]
- [S21] G. Seregin, A slightly supercritical condition of regularity of axisymmetric solutions to the Navier-Stokes equations, arXiv:2109.09344 (2021).
- [SS09] G. Seregin, V. Sverak, On Type I singularities of the local axisymmetric solutions of the Navier-Stokes equations, *Comm. Partial Differential Equations* 34 (2009), 171-201. [arXiv:0804.1803]
- [SSSZ12] G. Seregin, L. Silvestre, V. Sverak, A. Zlatoš, On divergence-free drifts, *J. Differential Equations* 252 (2012), 505-540. [arXiv:1010.6025]
- [SW60] J. T. Stuart, On the non-linear mechanics of wave disturbances in stable and unstable parallel flows. Part 1, *J. Fluid Mech.* 9 (1960), 353-370; J. Watson, Part 2, *J. Fluid Mech.* 9 (1960), 371-389.
- [SSF83] C. Sulem, P.-L. Sulem, H. Frisch, Tracing complex singularities with spectral methods, *J. Comput. Phys.* 50 (1983), 138-161.

- [T11] V. Theofilis, Global linear instability, *Annu. Rev. Fluid Mech.* 43 (2011), 319-352.
- [T16] T. Tao, Finite time blowup for an averaged three-dimensional Navier-Stokes equation, *J. Amer. Math. Soc.* 29 (2016), 601-674. [arXiv:1402.0290]
- [T98] T.-P. Tsai, On Leray's self-similar solutions of the Navier-Stokes equations satisfying local energy estimates, *Arch. Ration. Mech. Anal.* 143 (1998), 29-51.
- [VV20] A. Vasseur, M. Vishik, Blow-up solutions to 3D Euler are hydrodynamically unstable, *Comm. Math. Phys.* 378 (2020), 557-568. [arXiv:1908.05766]
- [V96] M. M. Vishik, Spectrum of small oscillations of an ideal fluid and Lyapunov exponents, *J. Math. Pures Appl.* 75 (1996), 531-558.
- [WLGB23] Y. Wang, C.-Y. Lai, J. Gomez-Serrano, T. Buckmaster, Asymptotic self-similar blow-up profile for three-dimensional axisymmetric Euler equations using neural networks, *Phys. Rev. Lett.* 130 (2023), 244002. [arXiv:2201.06780]
- [W25] Y. Wang, M. Bennani, J. Martens, S. Racaniere, S. Blackwell, A. Matthews, S. Nikolov, G. Cao-Labora, D. S. Park, M. Arjovsky, D. Worrall, C. Qin, F. Alet, B. Kozlovskii, N. Tomasev, A. Davies, P. Kohli, T. Buckmaster, B. Georgiev, J. Gomez-Serrano, R. Jiang, C.-Y. Lai, Discovery of unstable singularities, arXiv:2509.14185 (2025).

Every entry above was checked against the arXiv abstract page, Crossref, or the journal page in two independent passes; the Found. Comput. Math. volume and page numbers of [H23a] and [H23b] and the English-edition volume of [LZZ17] remain to be filled from the publishers' pages.